
Course Outline

1. Introduction
 2. Neuronal Signaling
 3. Visual System
 4. Neural Information Processing
 5. Decision making
 6. Multi-modal Sensory Systems
 7. Hippocampus, Spatial Cognition & Memory
 8. Value-based Decision Making
 9. Motor Systems
 10. Learning & Plasticity
 11. Attention and More
 12. Project Presentation
 13. Final Q&A
-

系统和计算神经科学导论

十一、注意和相关认知过程

Attention and More

中国科学院脑科学与智能技术卓越创新中心（神经所） 杨天明 TYANG@ION.AC.CN

-
- ❖ **Overview**
 - ❖ **Visual Attention**
 - ❖ **Attention Modulation of Visual Processing**
 - ❖ **Top-down vs Bottom-up**
 - ❖ **And more...**
-



❖ Overview

❖ Visual Attention

❖ Attention Modulation of
Visual Processing

❖ Top-down vs Bottom-up

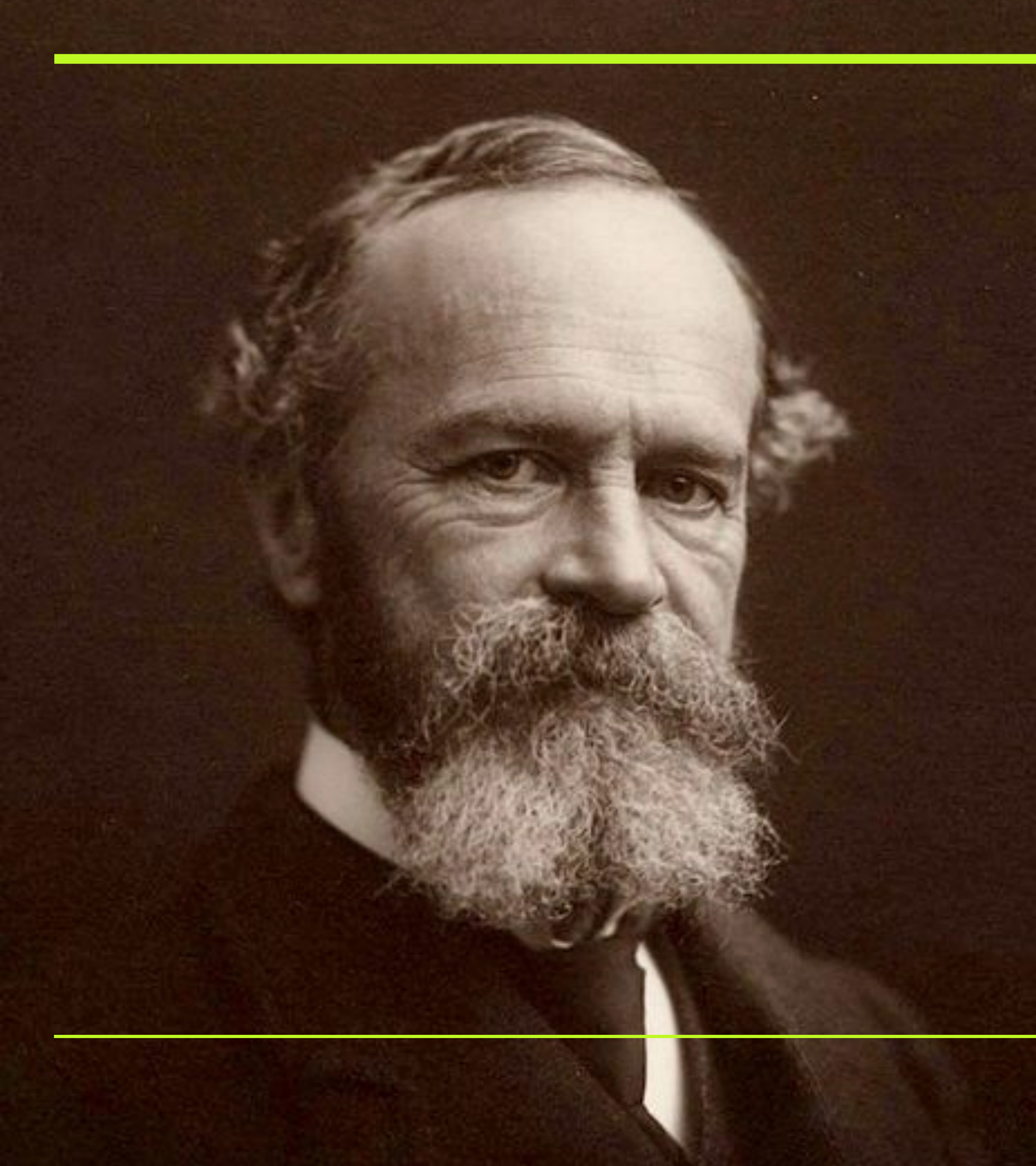
❖ And more...



Quirkology Channel

THE COLOUR CHANGING CARD TRICK

www.RichardWiseman.com



What is attention?

Everyone knows what attention is... the taking possession by the mind, in clear and vivid form, of one out of what seem several simultaneously possible objects or trains of thought. Focalization, concentration, of consciousness are of its essence. It implies withdrawal from some things in order to deal effectively with others.

-- William James

-
- ❖ Overview
 - ❖ **Visual Attention**
 - ❖ Attention Modulation of Visual Processing
 - ❖ Top-down vs Bottom-up
 - ❖ And more...
-



Visual Attention

- Overt Attention
to direct the gaze toward an object
- Covert Attention
to look out of the corner of your eye

Stay calm, be cool...





Top-down



Bottom-up

Visual Attention

- Voluntary Attention
top-down
- Involuntary Attention
bottom-up



Visual Attention

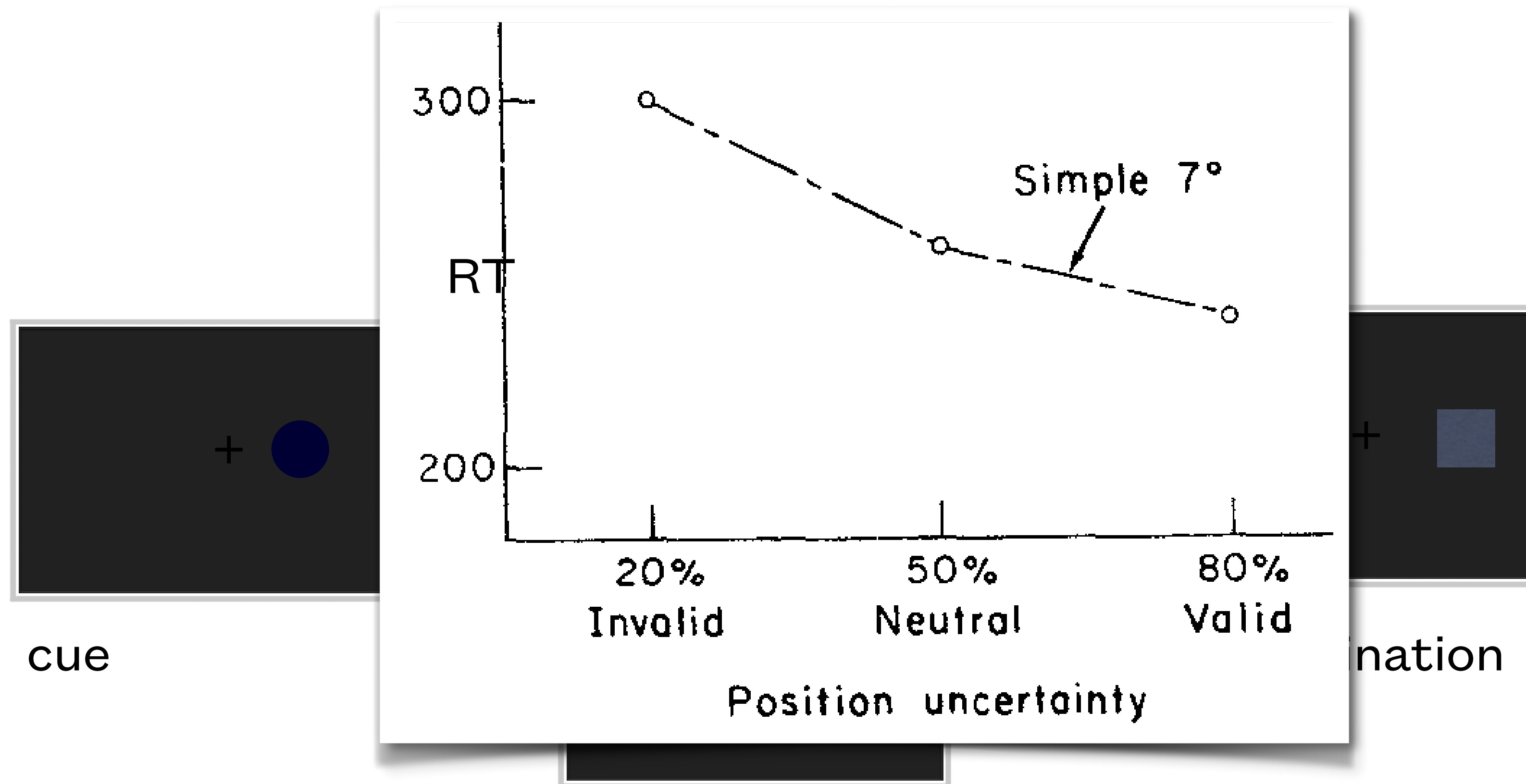
- Spatial Attention
- Feature Attention



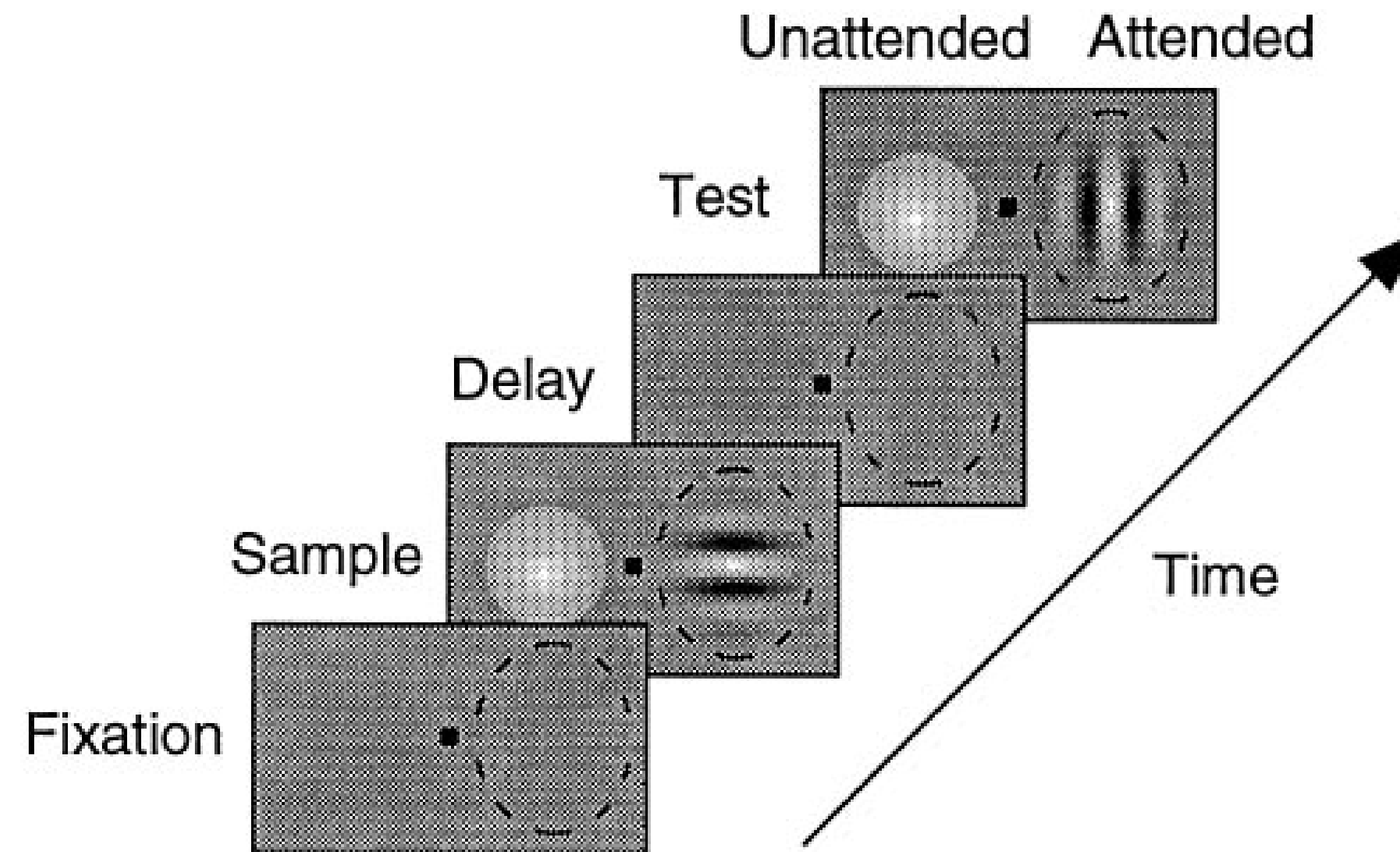
-
- ❖ Overview
 - ❖ Visual Attention
 - ❖ **Attention Modulation of Visual Processing**
 - ❖ Top-down vs Bottom-up
 - ❖ And more...
-



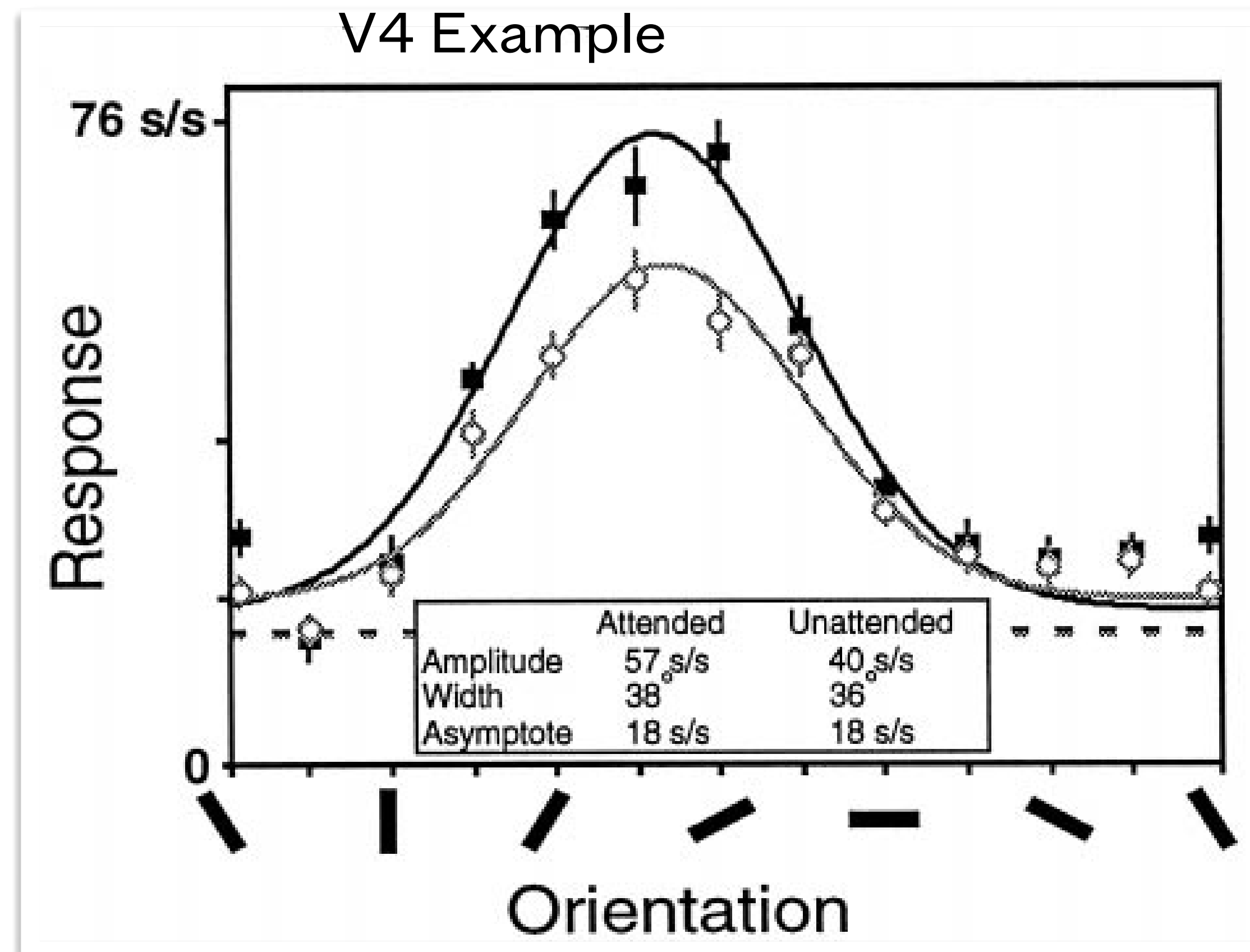
Posner Paradigm



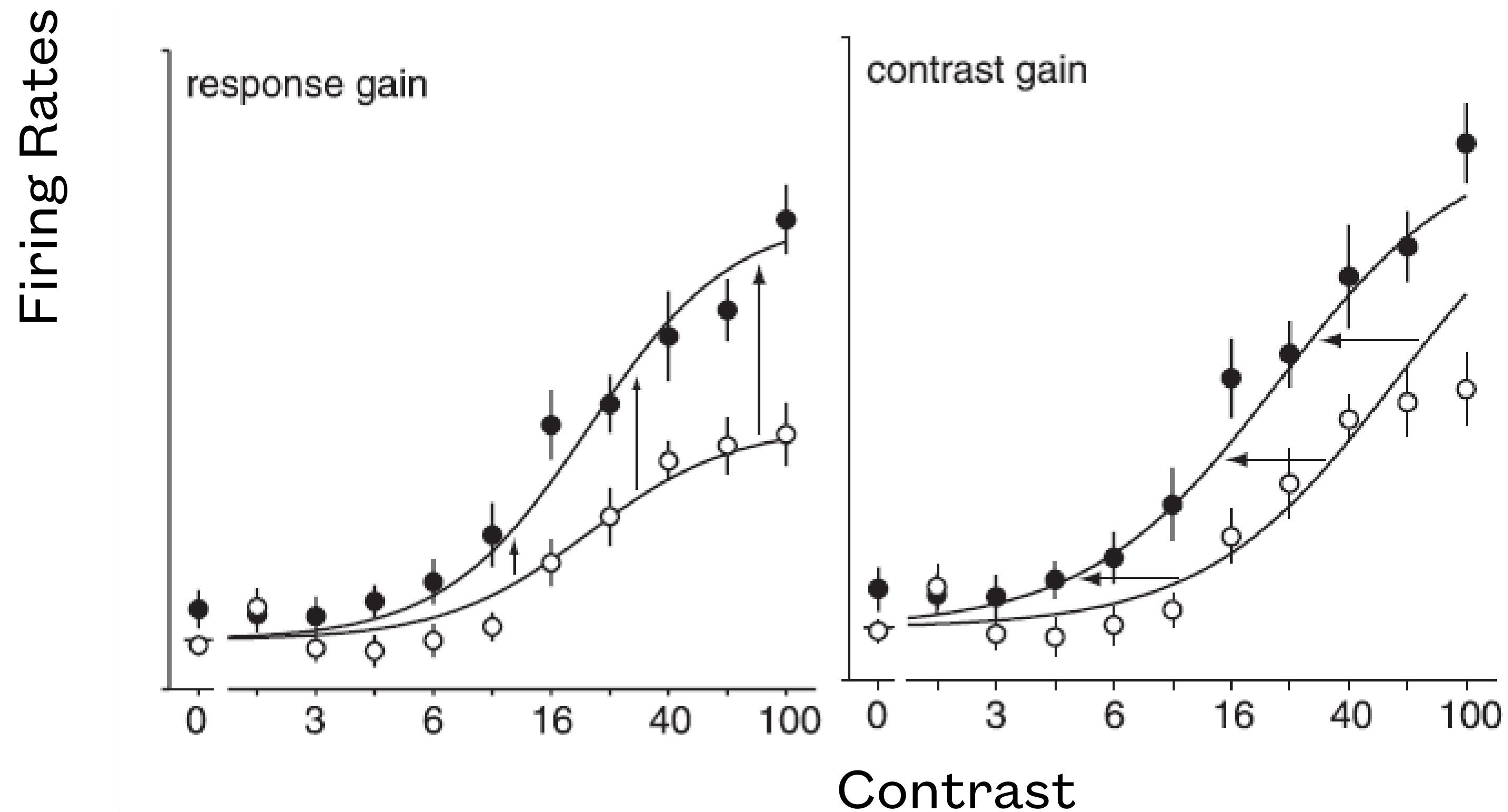
Attention effects on the Orientation Tuning



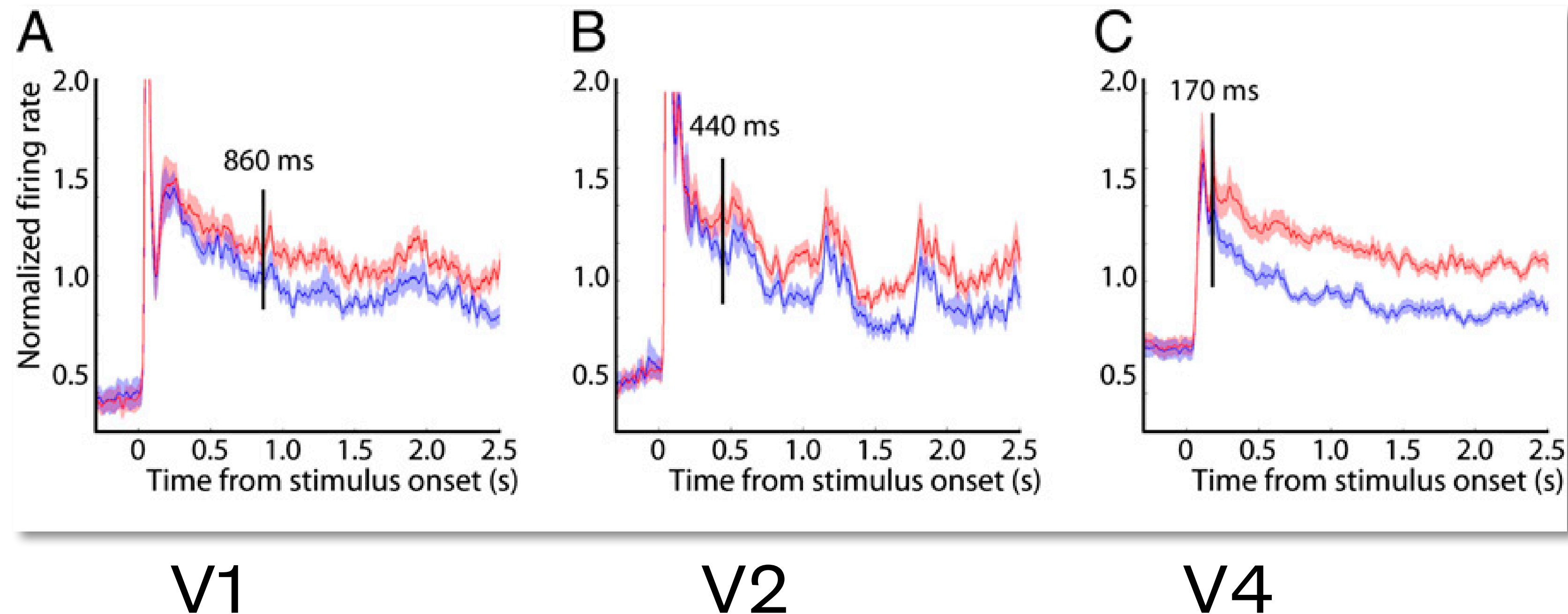
Attention effects on the orientation tuning



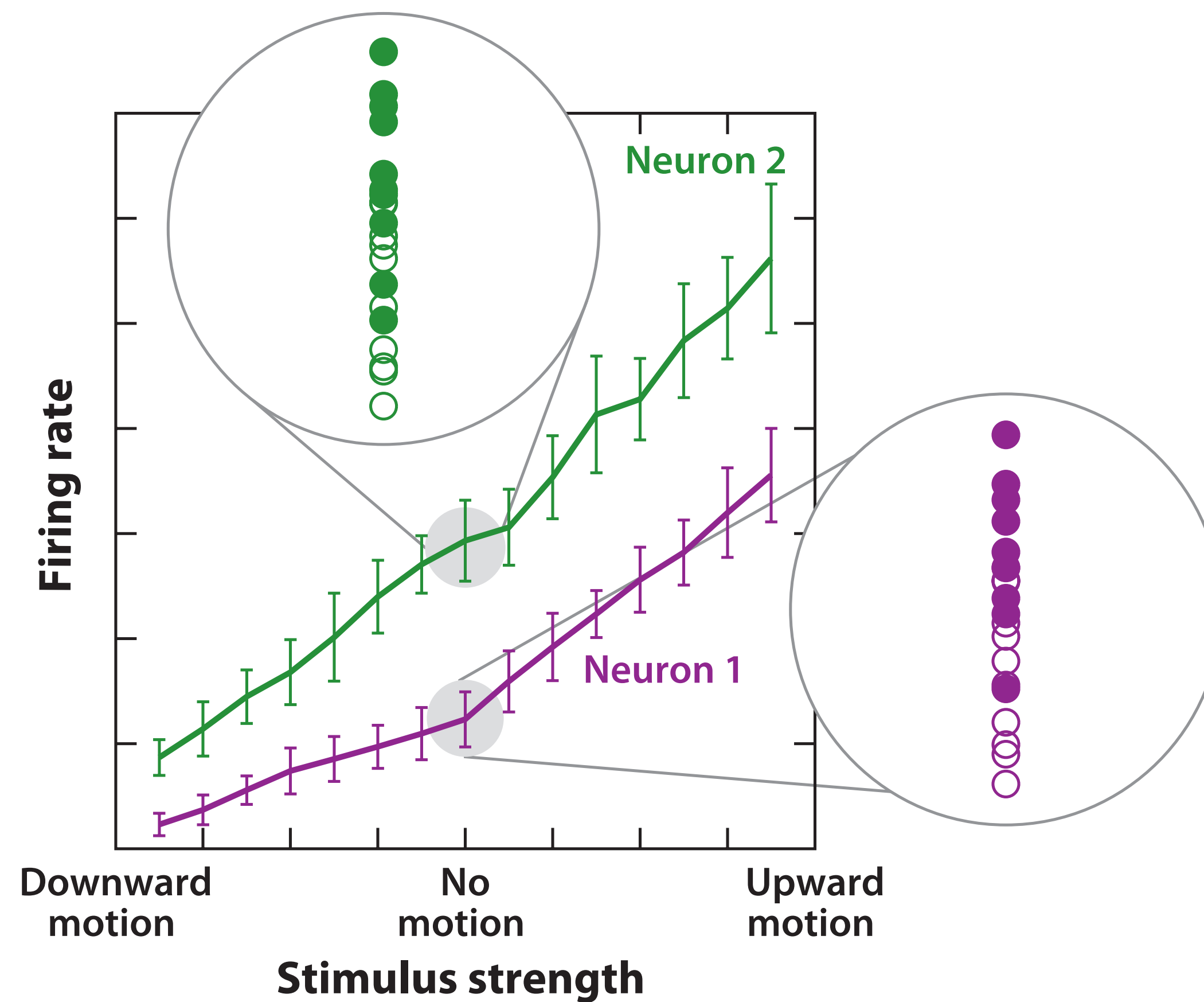
Contrast or Response Gain?



Attention effects in different visual areas



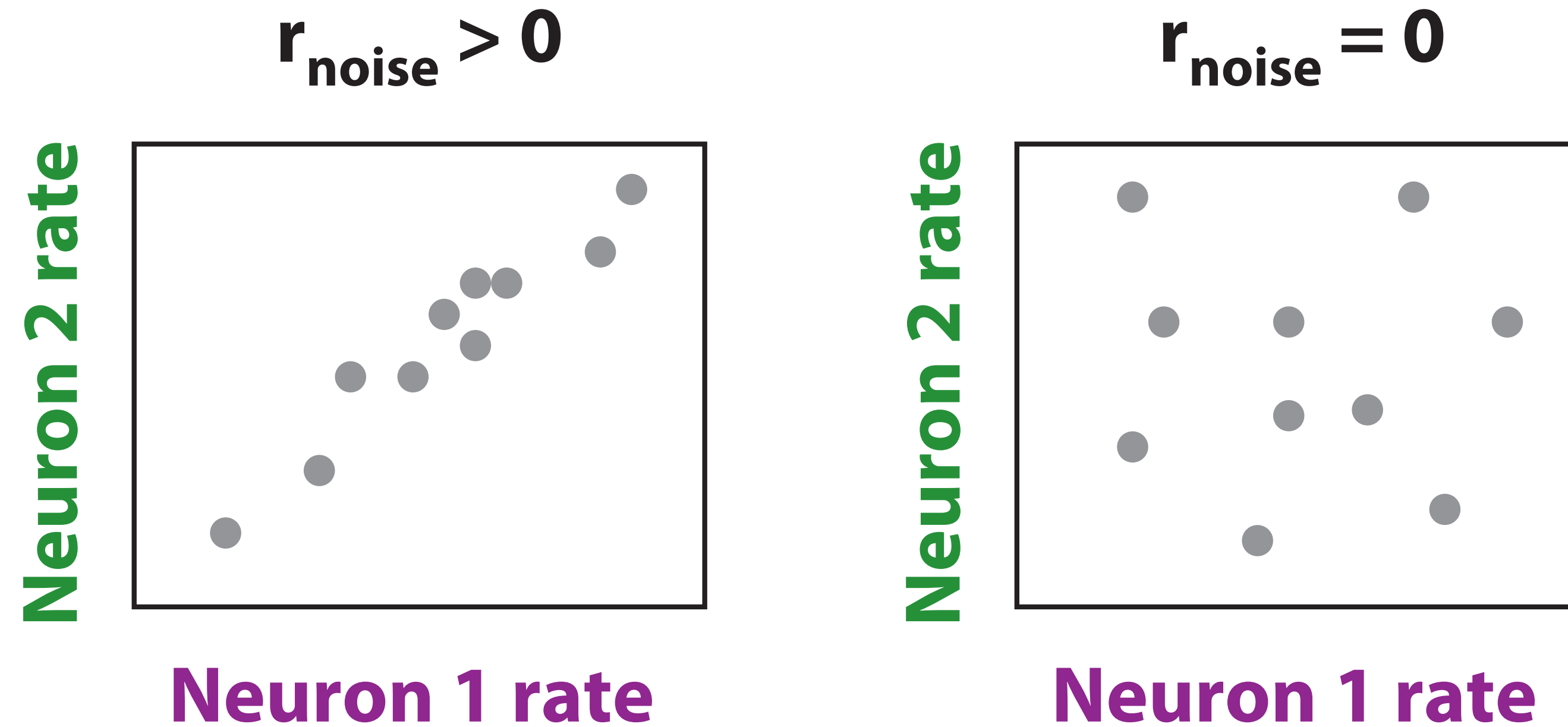
Pairwise correlation



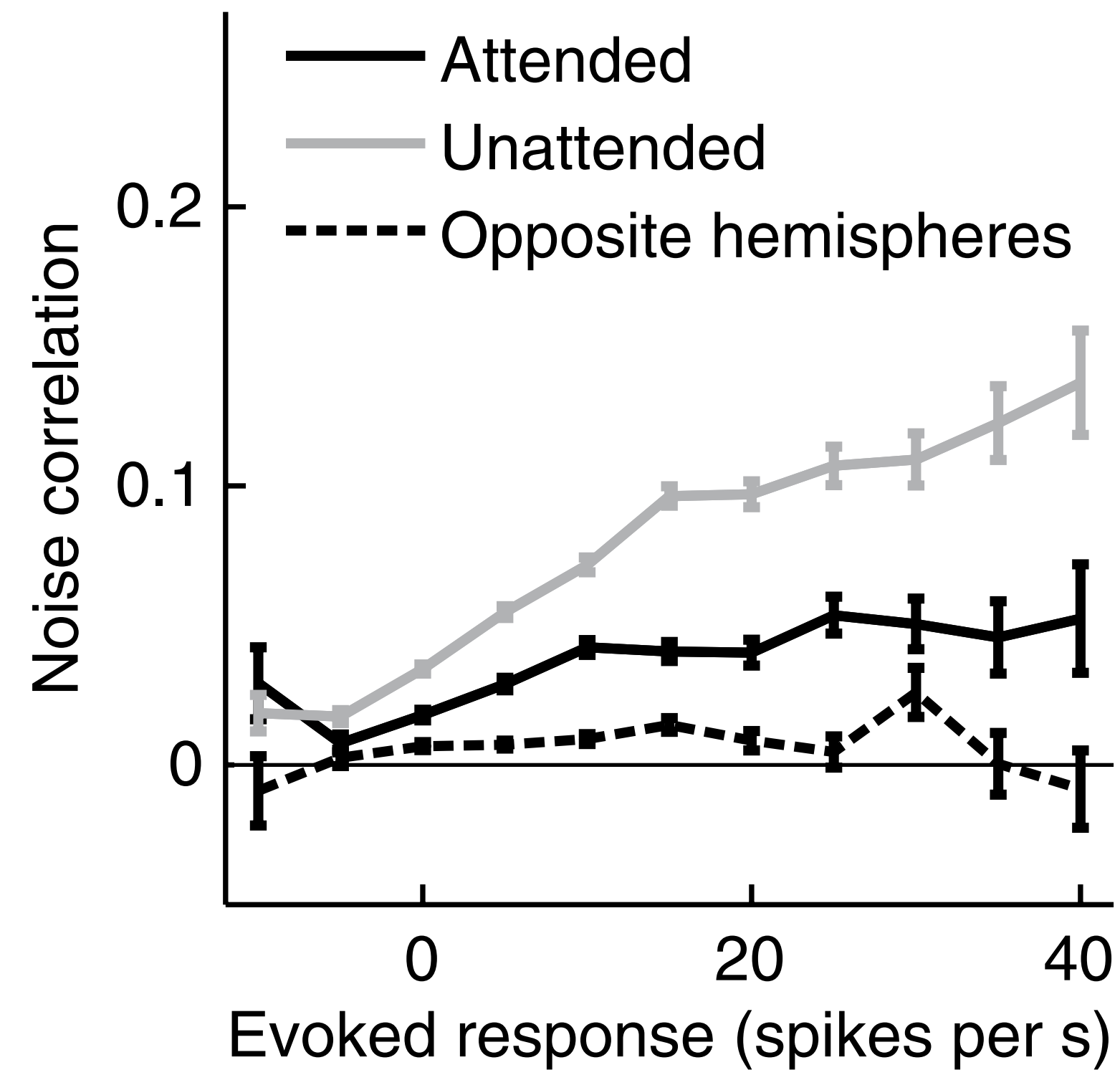
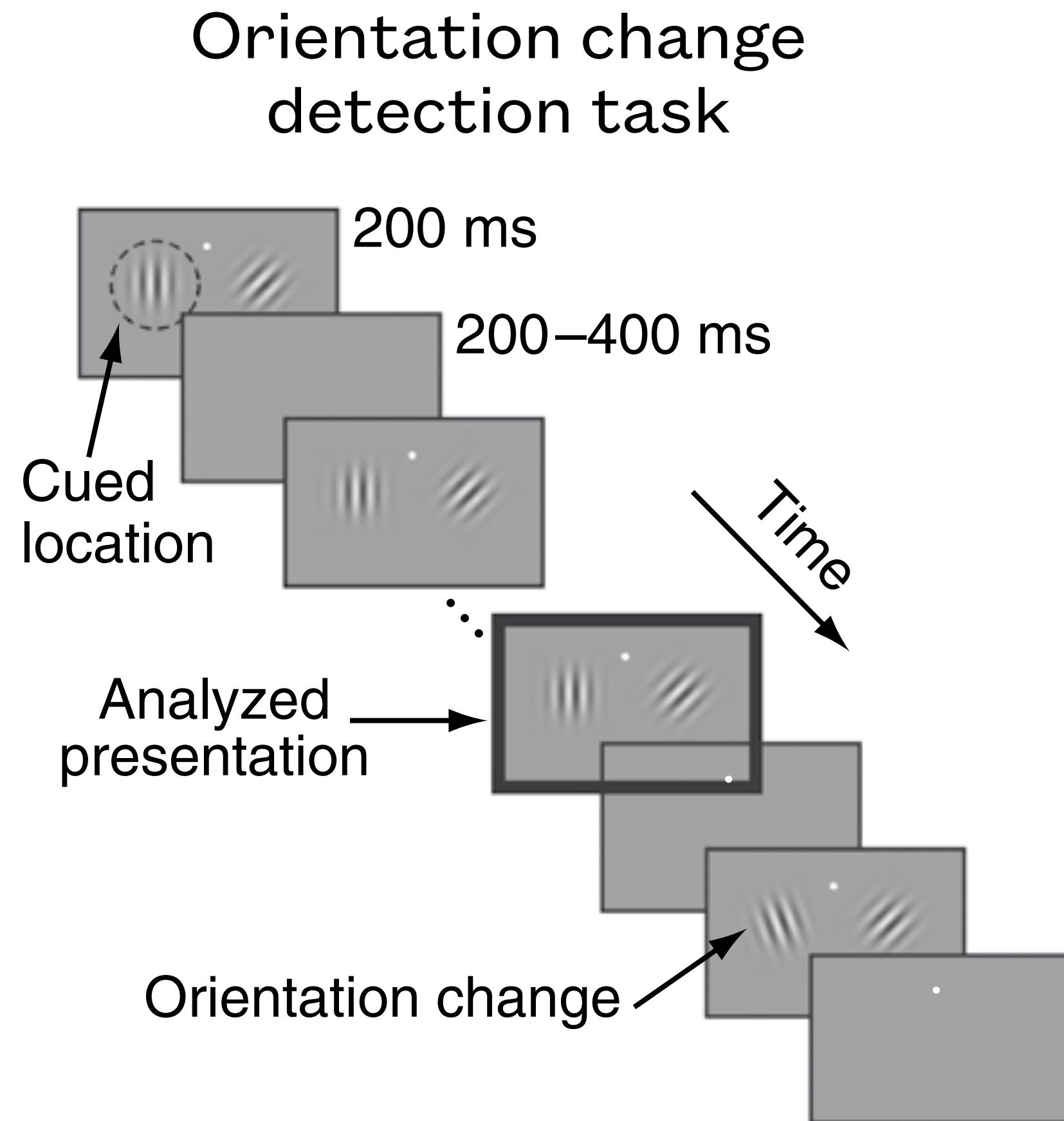
Signal Correlation

Pairwise correlation

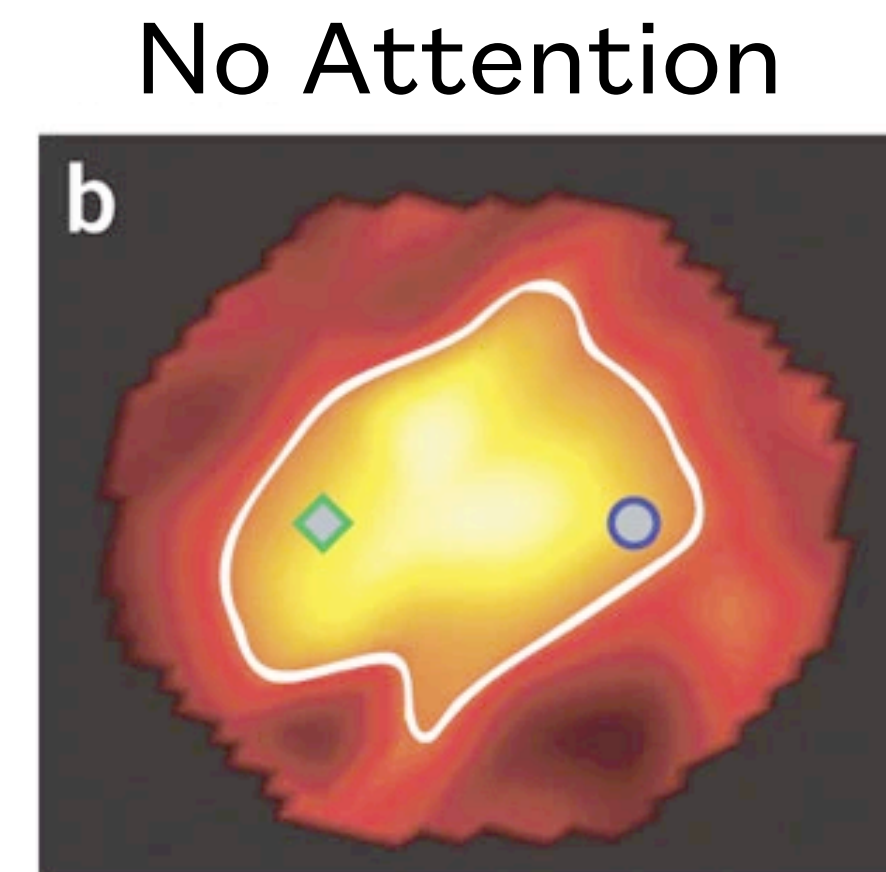
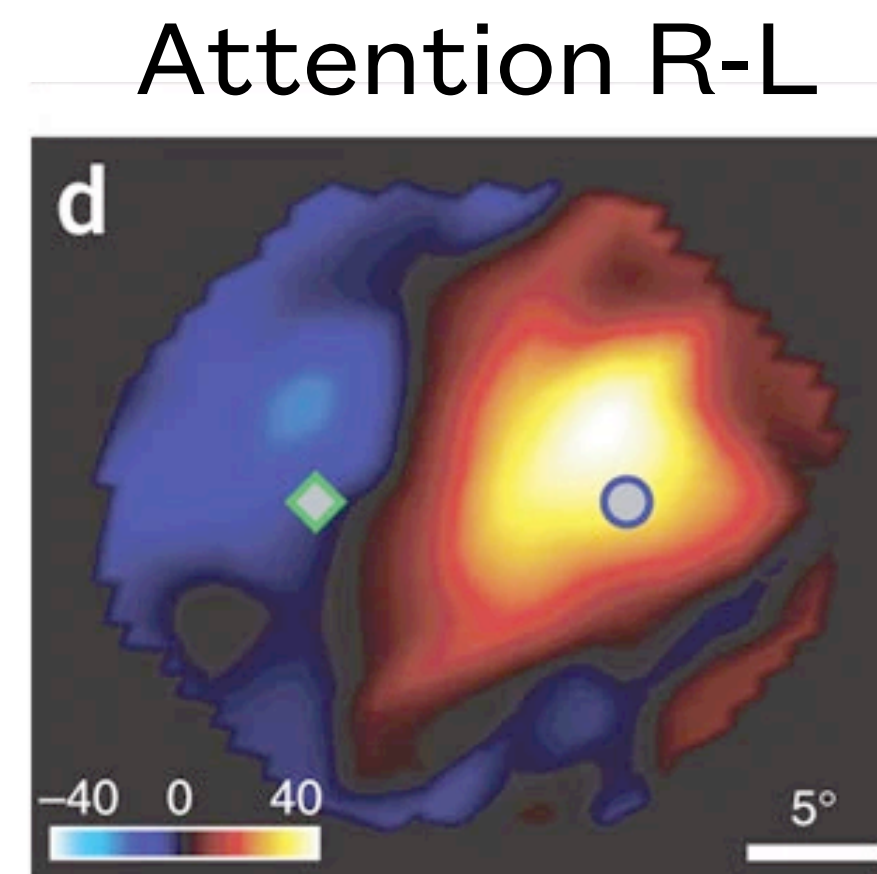
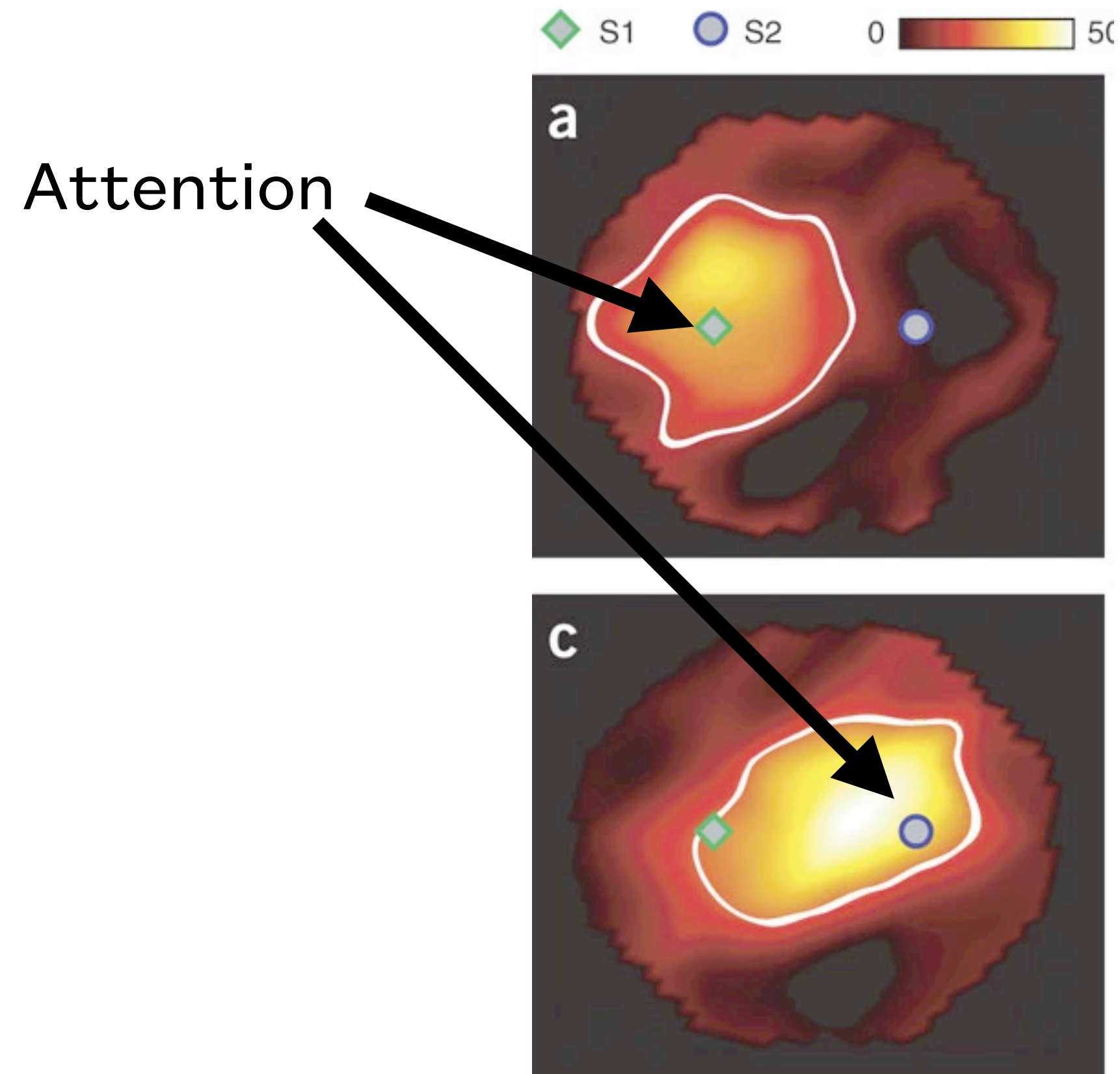
Noise correlation



Attention changes noise correlation



Receptive field changes

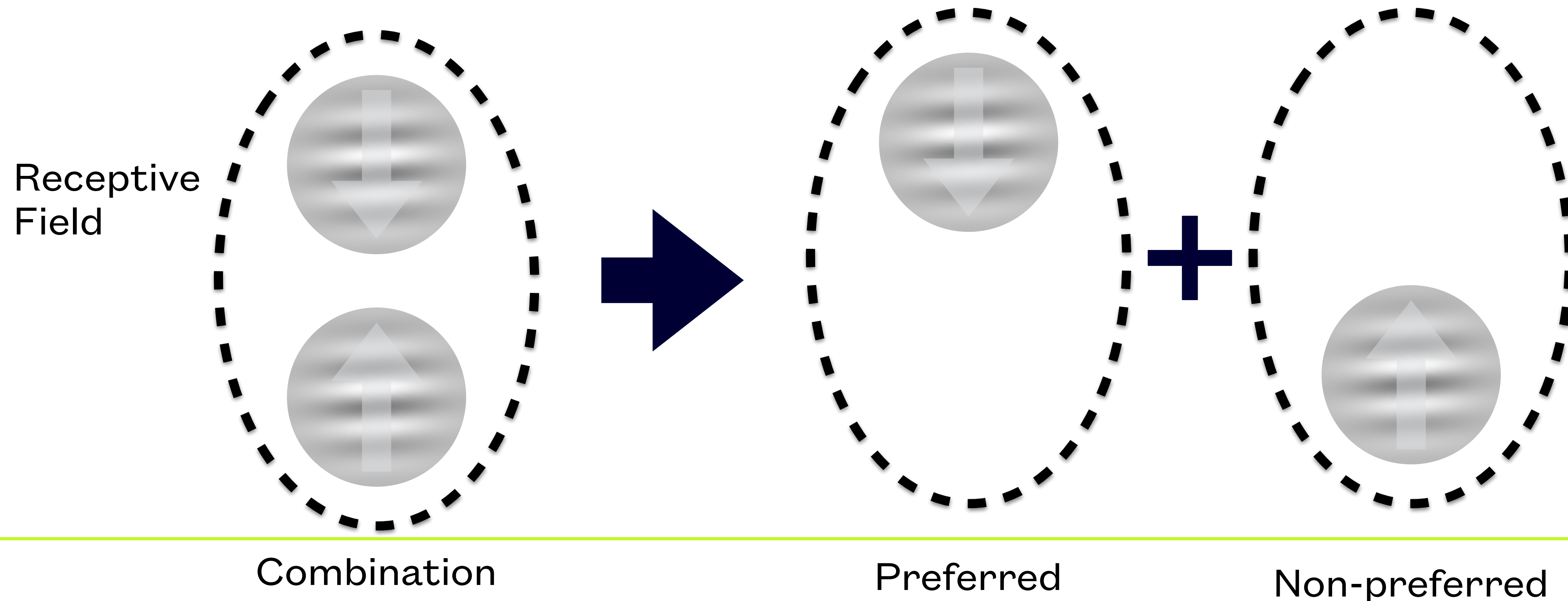


Attention changes visual responses

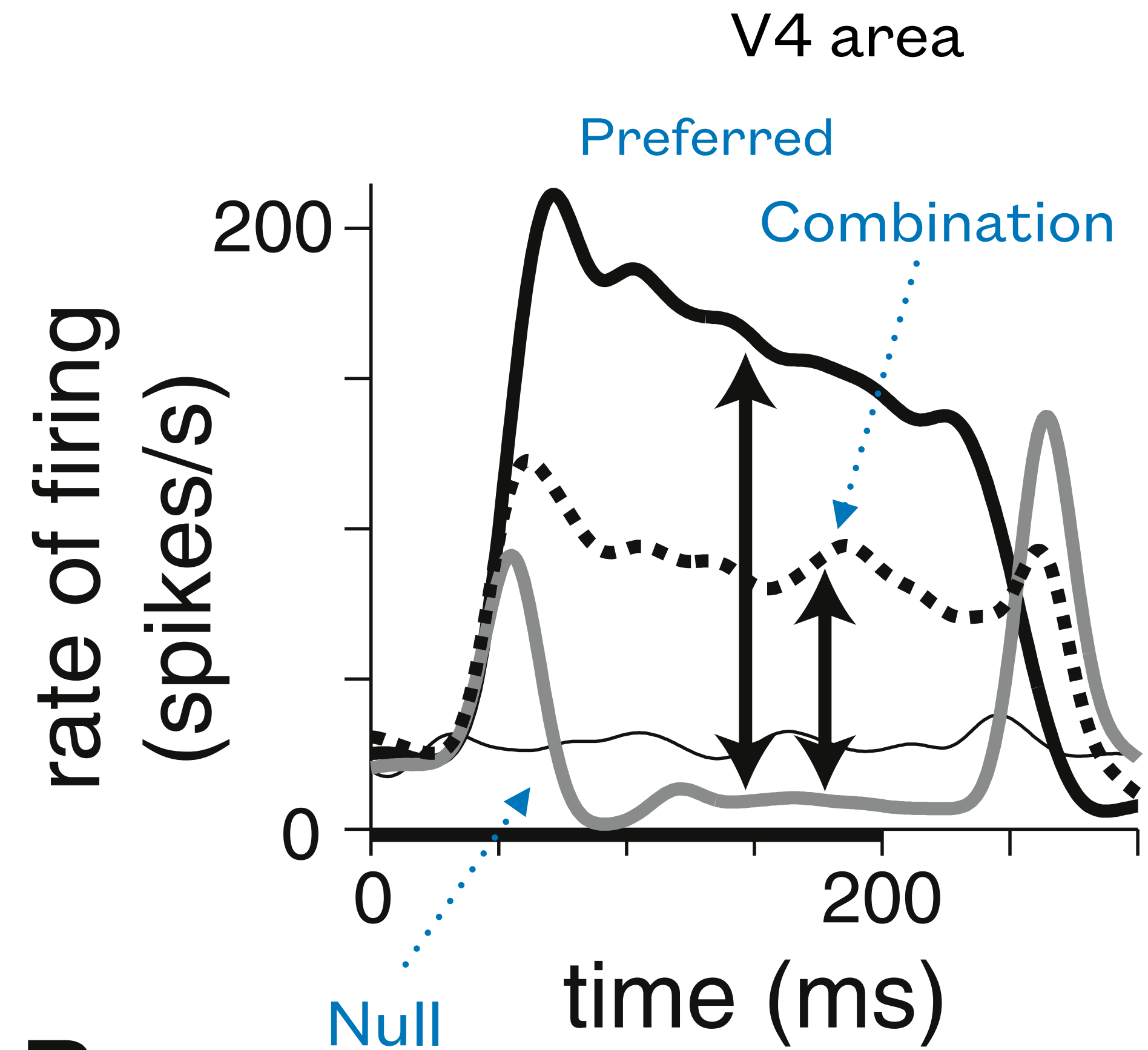
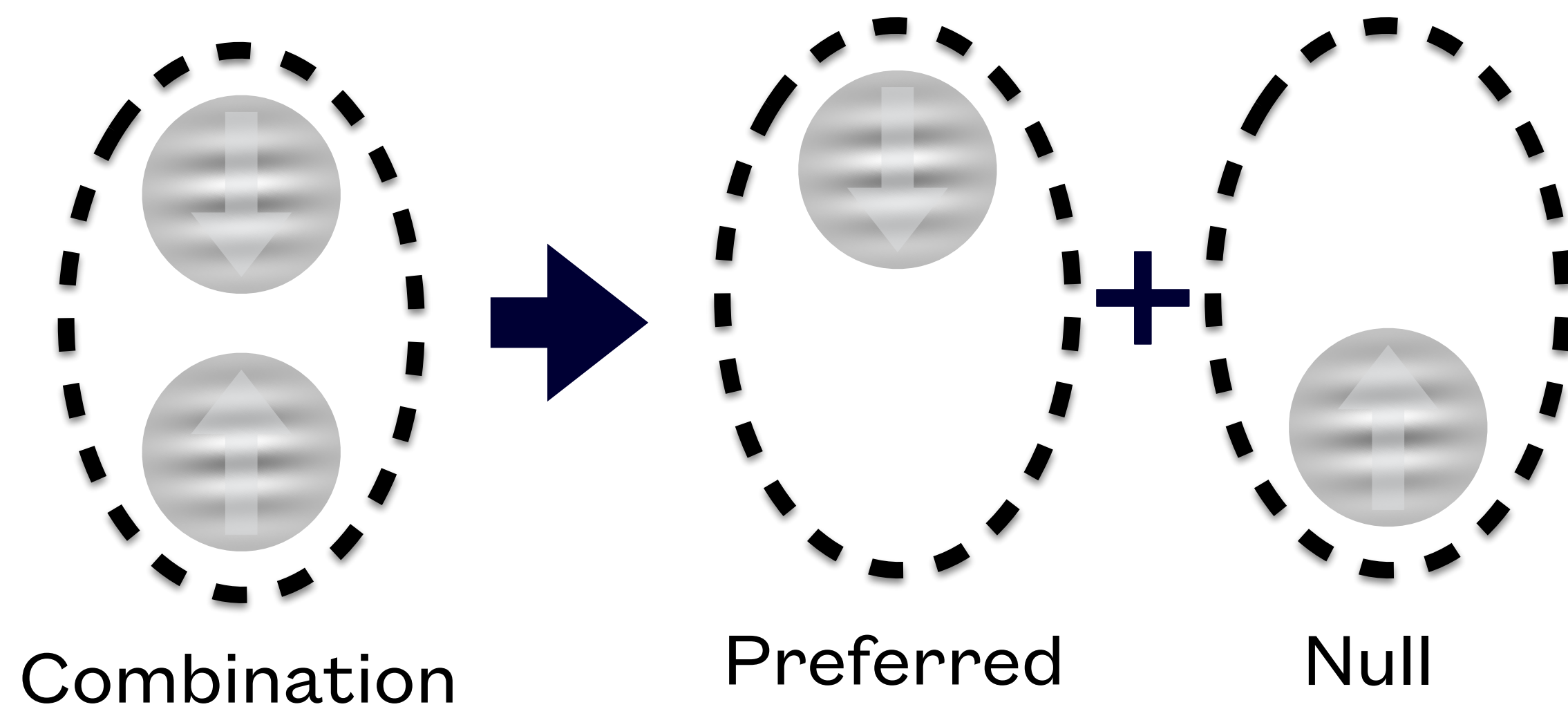
- Firing rate changes
 - modest: 5-30%
 - Gain change or contrast change
 - Larger in late stages of visual processing
 - Lower noise correlation
 - Receptive field changes
-

Attention and sensory normalization

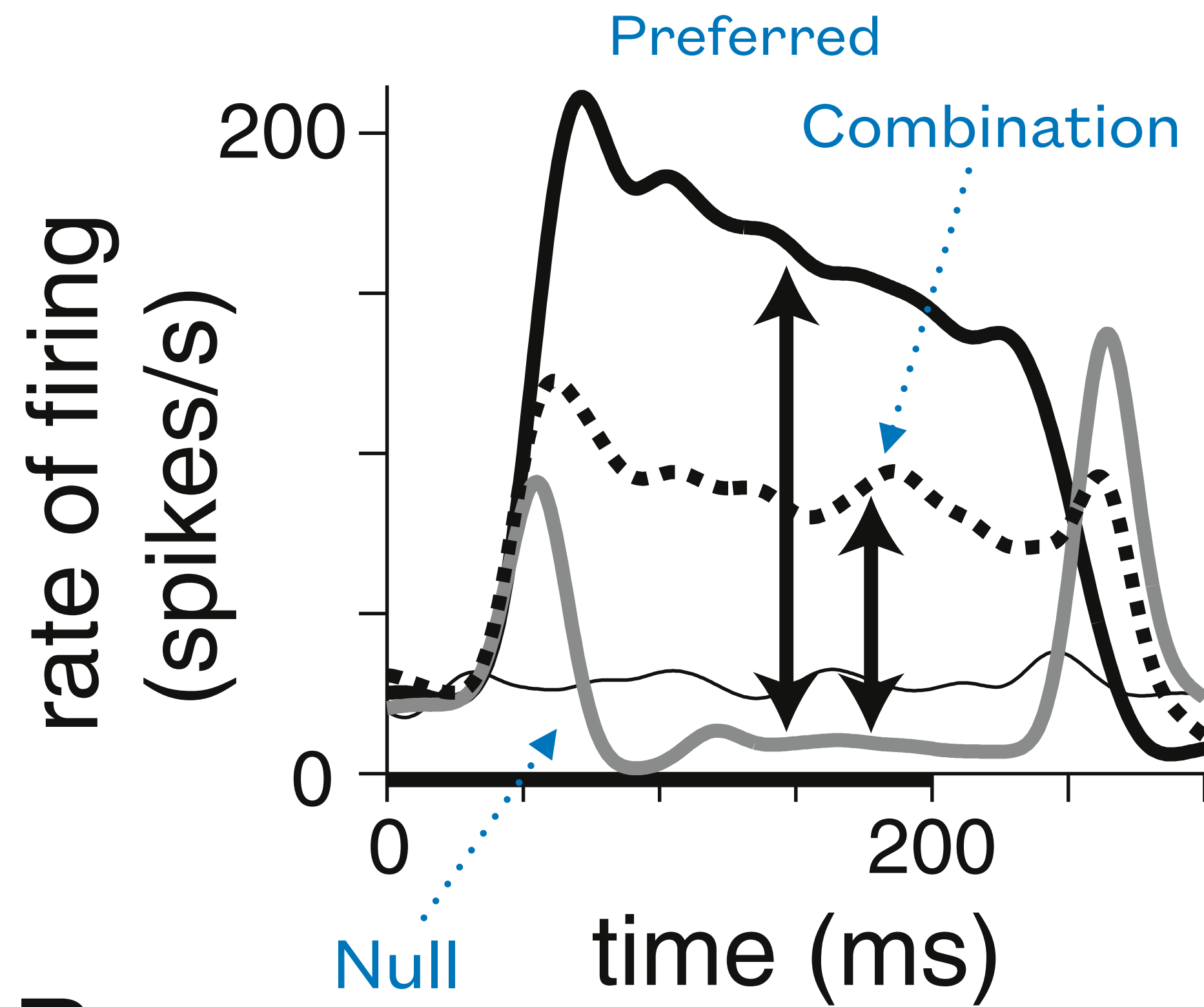
- Normalization: not a specific mathematical operation, rather than a nonlinear summation of excitatory and suppressive inputs from multiple sources.



Attention and sensory normalization



Attention and sensory normalization



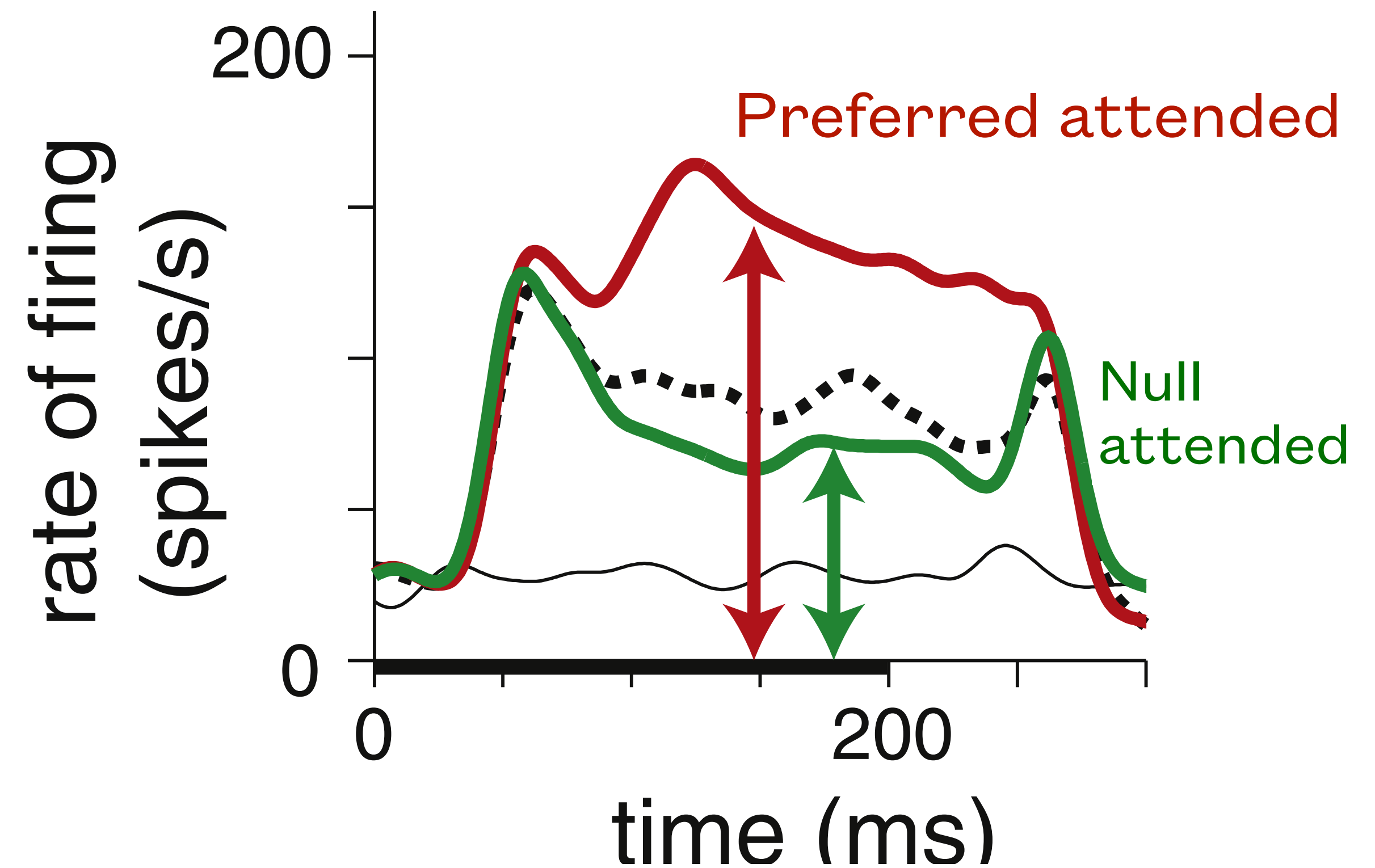
$$R_i(t) = K \frac{A_i(t)}{\sigma^2 + \sum_j A_j(t)}$$

Divisive Normalization

Heeger, 1993

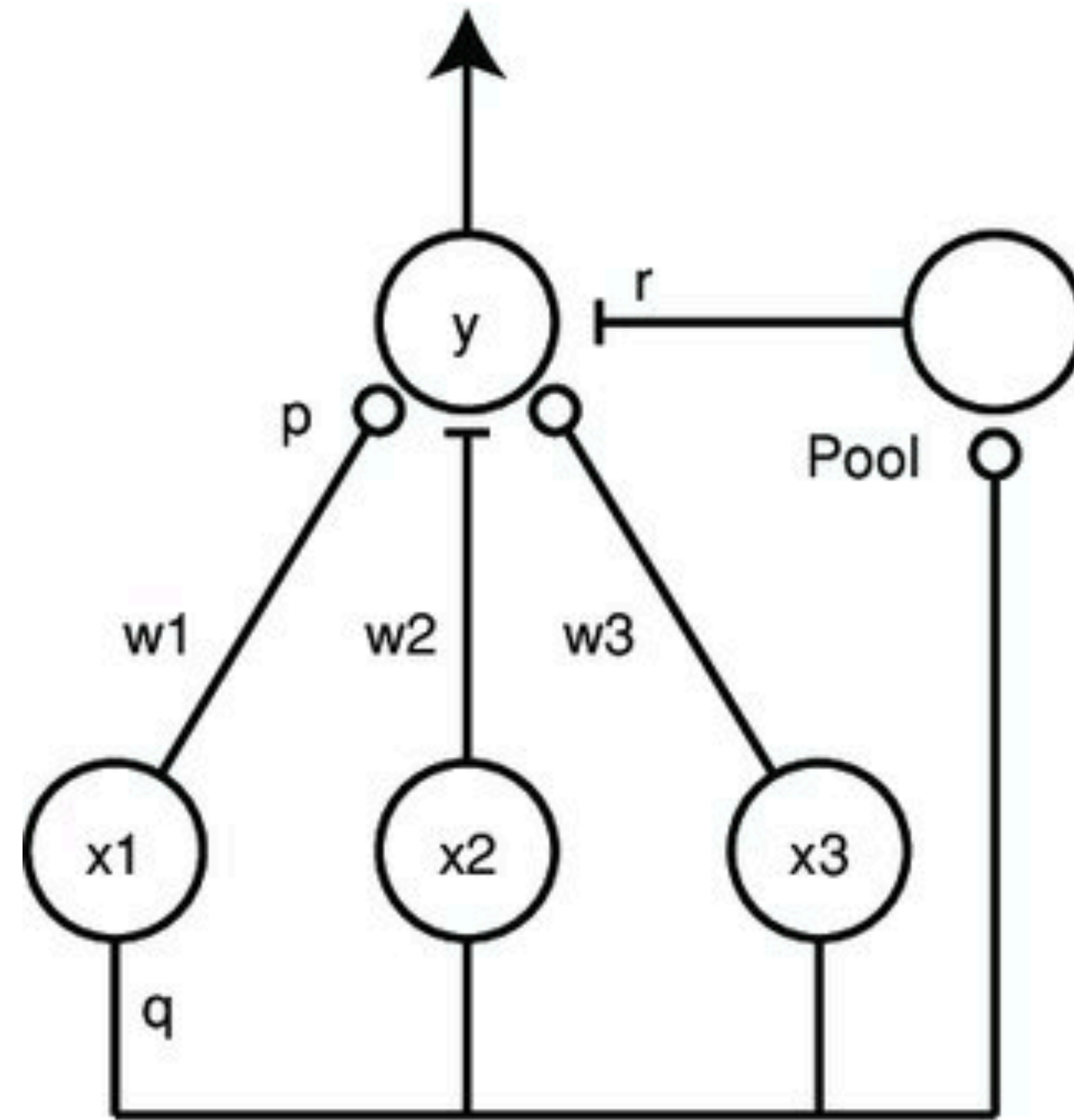
Attention and sensory normalization

$$R_{P,N} = \frac{C_P L_P + C_N L_N}{C_P + C_N + \sigma},$$

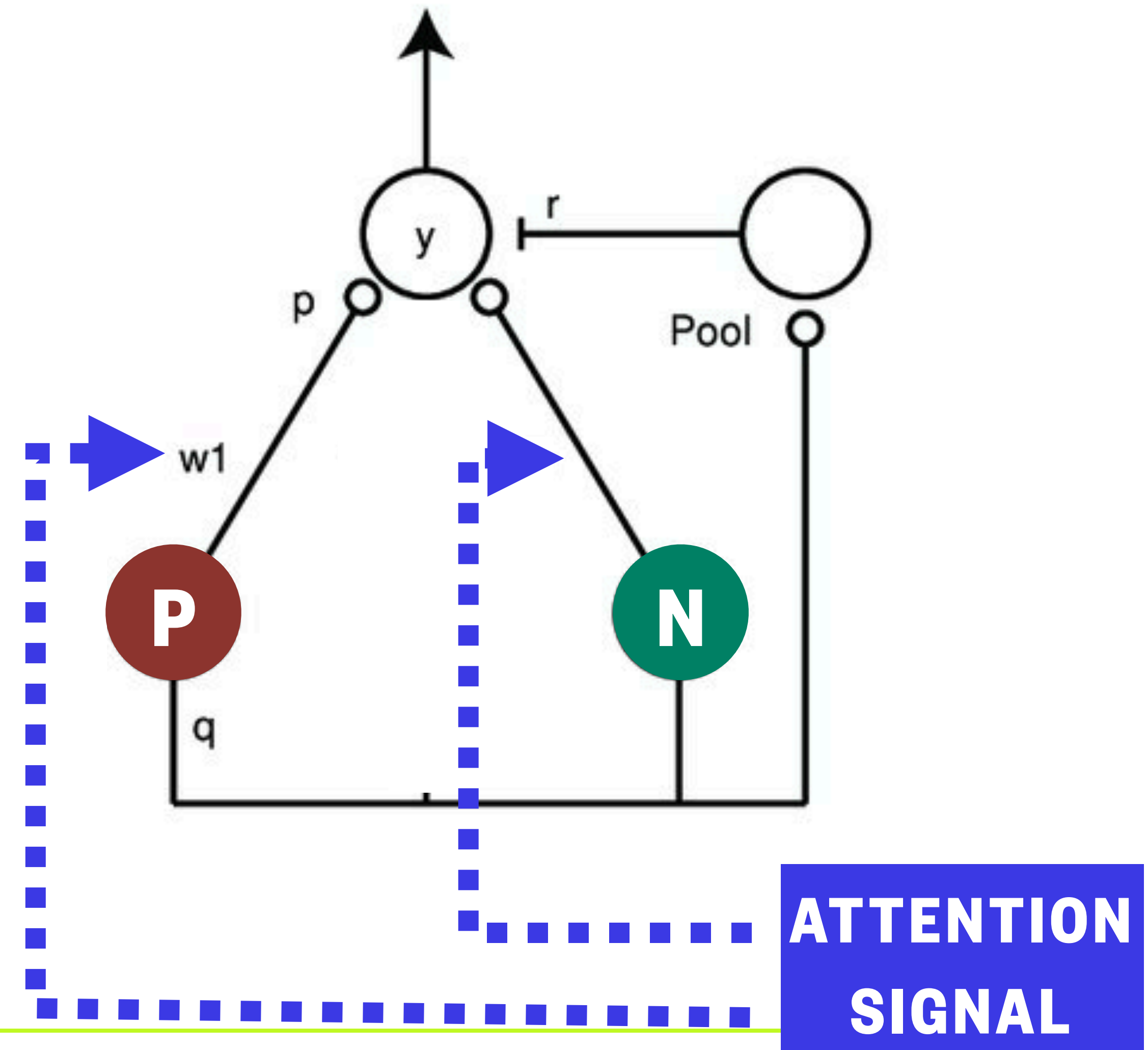
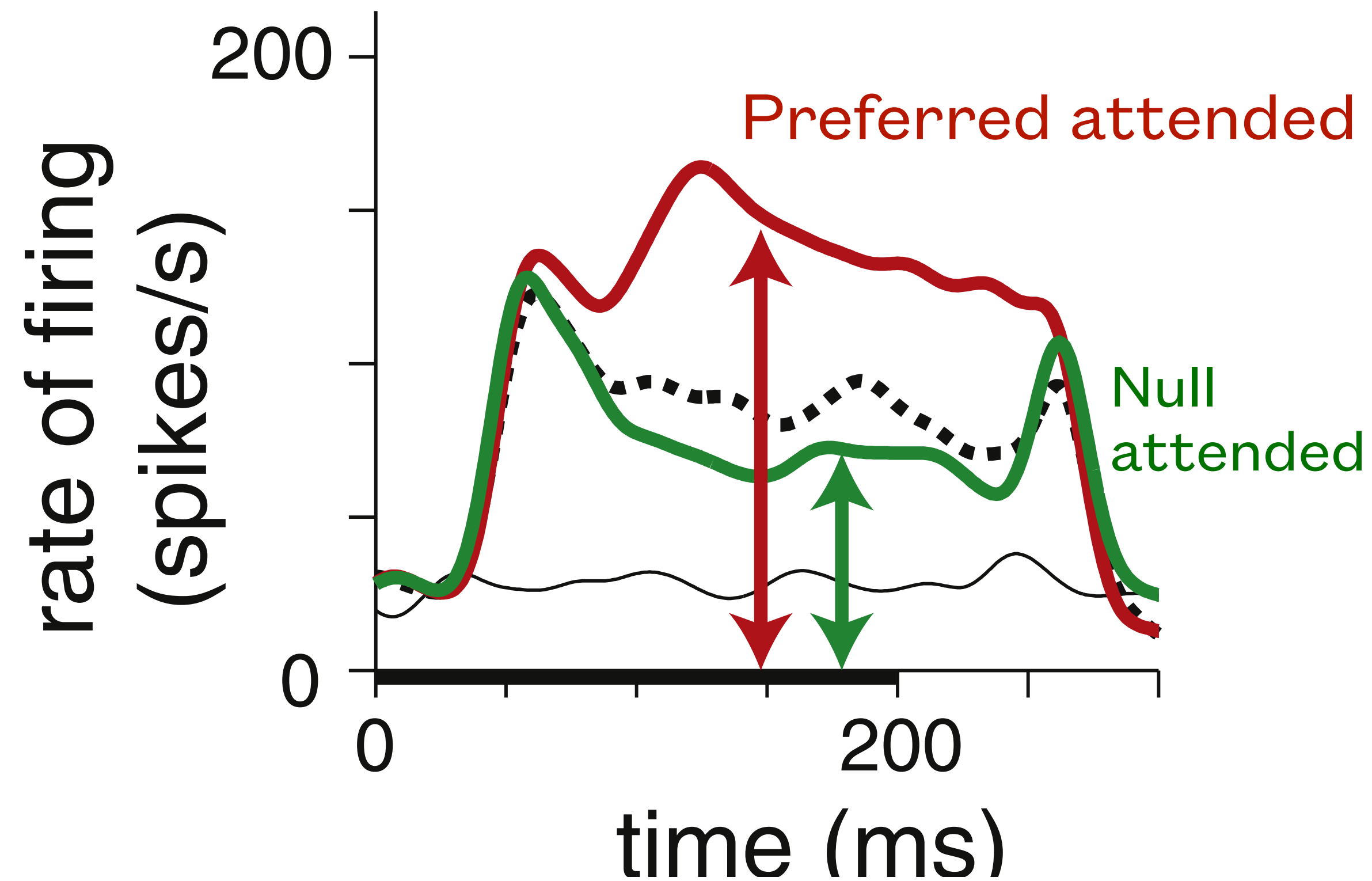


Circuit Model for Normalization

$$R_i(t) = K \frac{A_i(t)}{\sigma^2 + \sum_j A_j(t)}$$



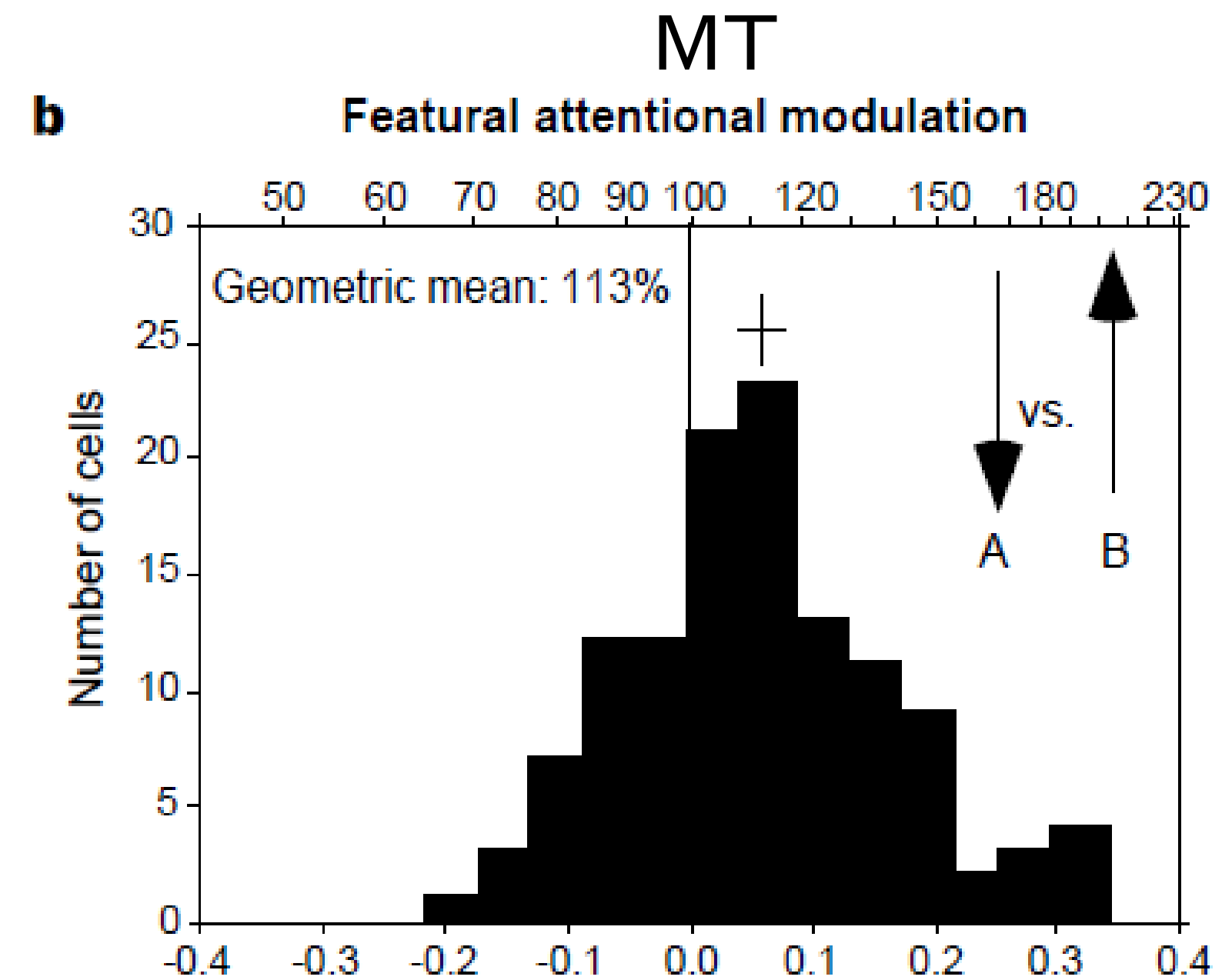
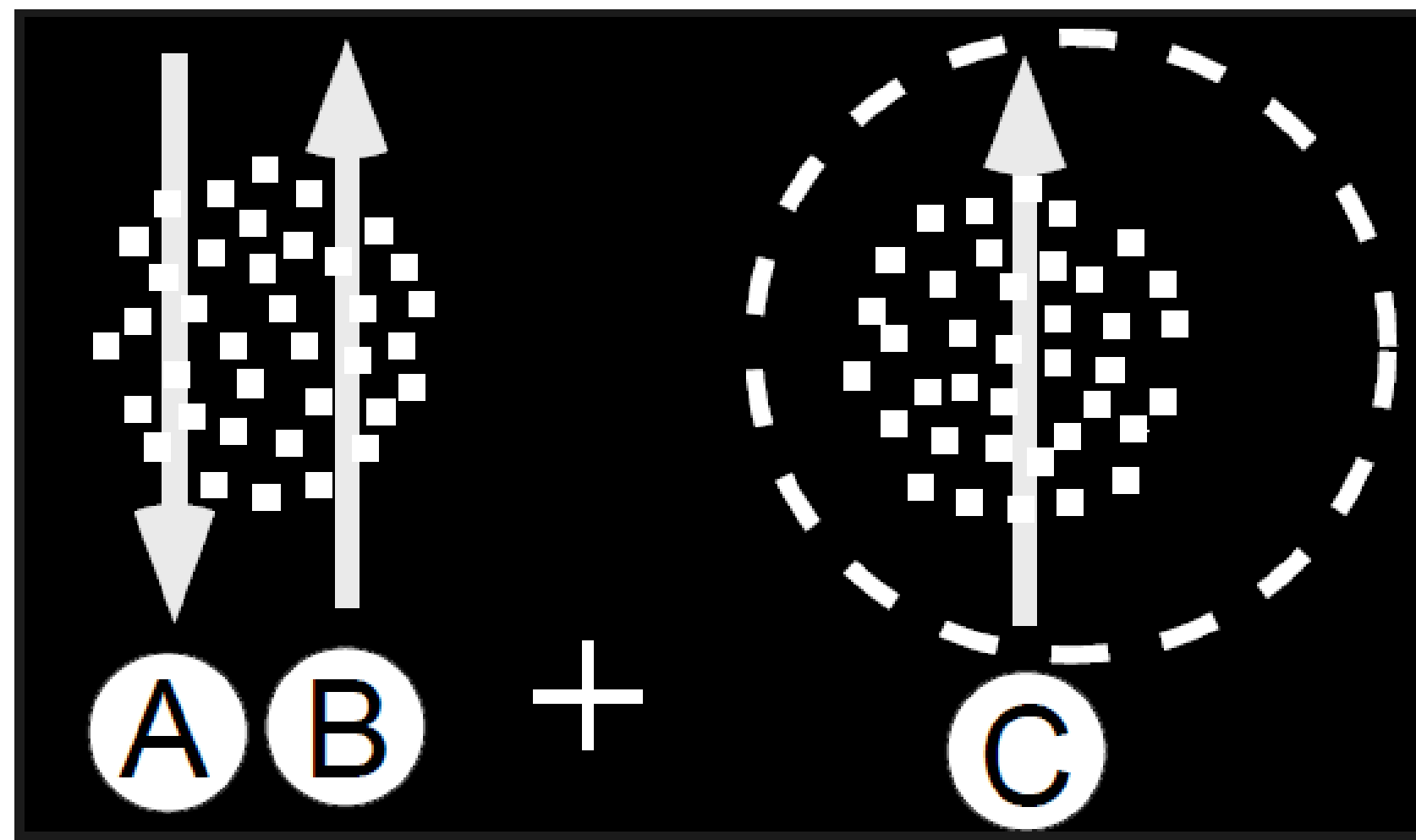
Circuit Model for Normalization



Attention and sensory normalization

- Attention does not alter the normalization mechanism.
 - It alters only the inputs to the normalization.
 - Normalization is a separate stage of processing that comes after the attention-related modulation.
-

Feature-based attention

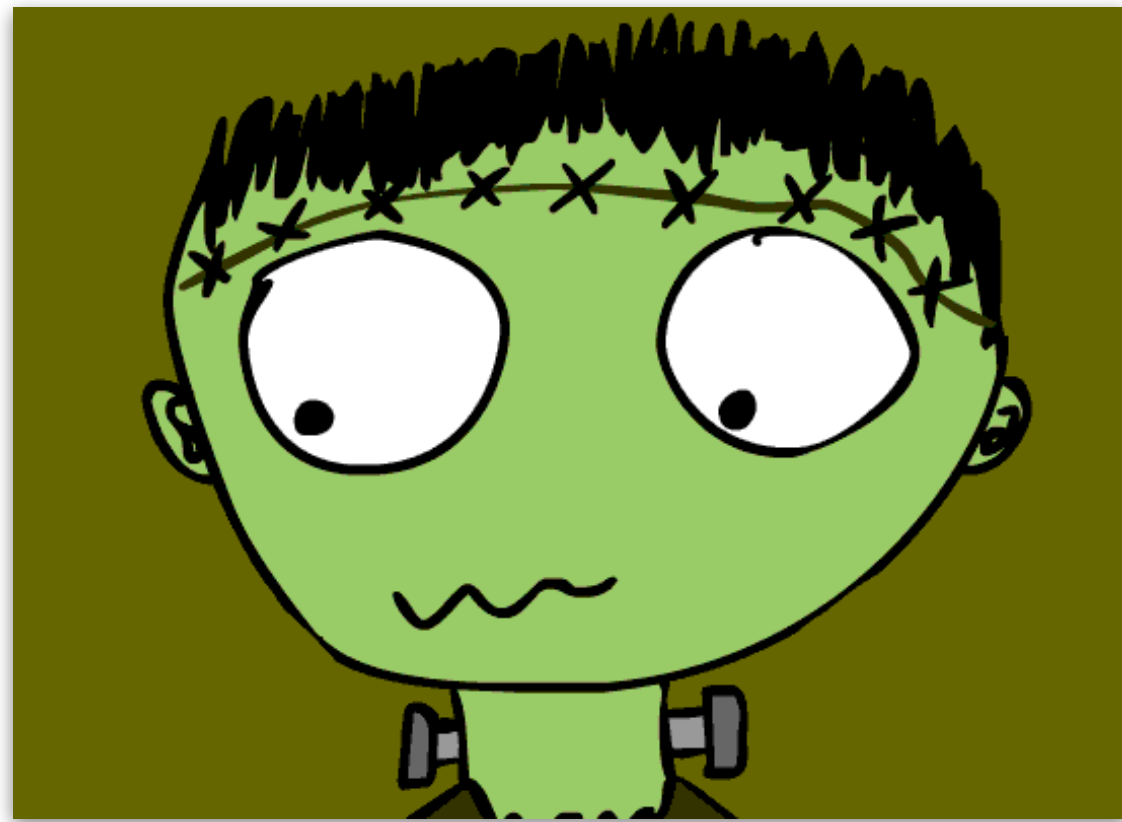


Treue and Martínez-Trujillo, 1999

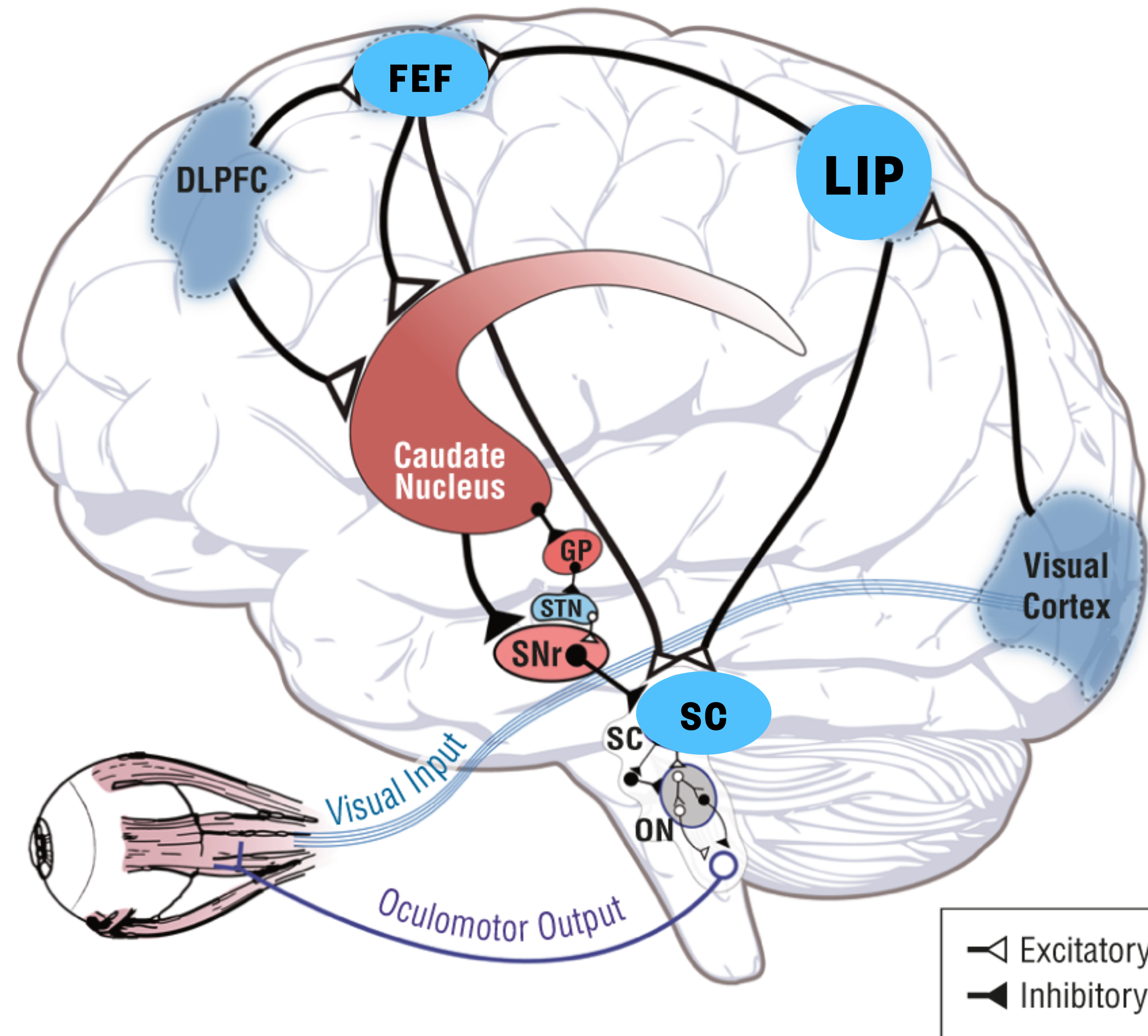
-
- ❖ Overview
 - ❖ Visual Attention
 - ❖ Attention Modulation of Visual Processing
 - ❖ **Top-down vs Bottom-up**
 - ❖ And more...
-



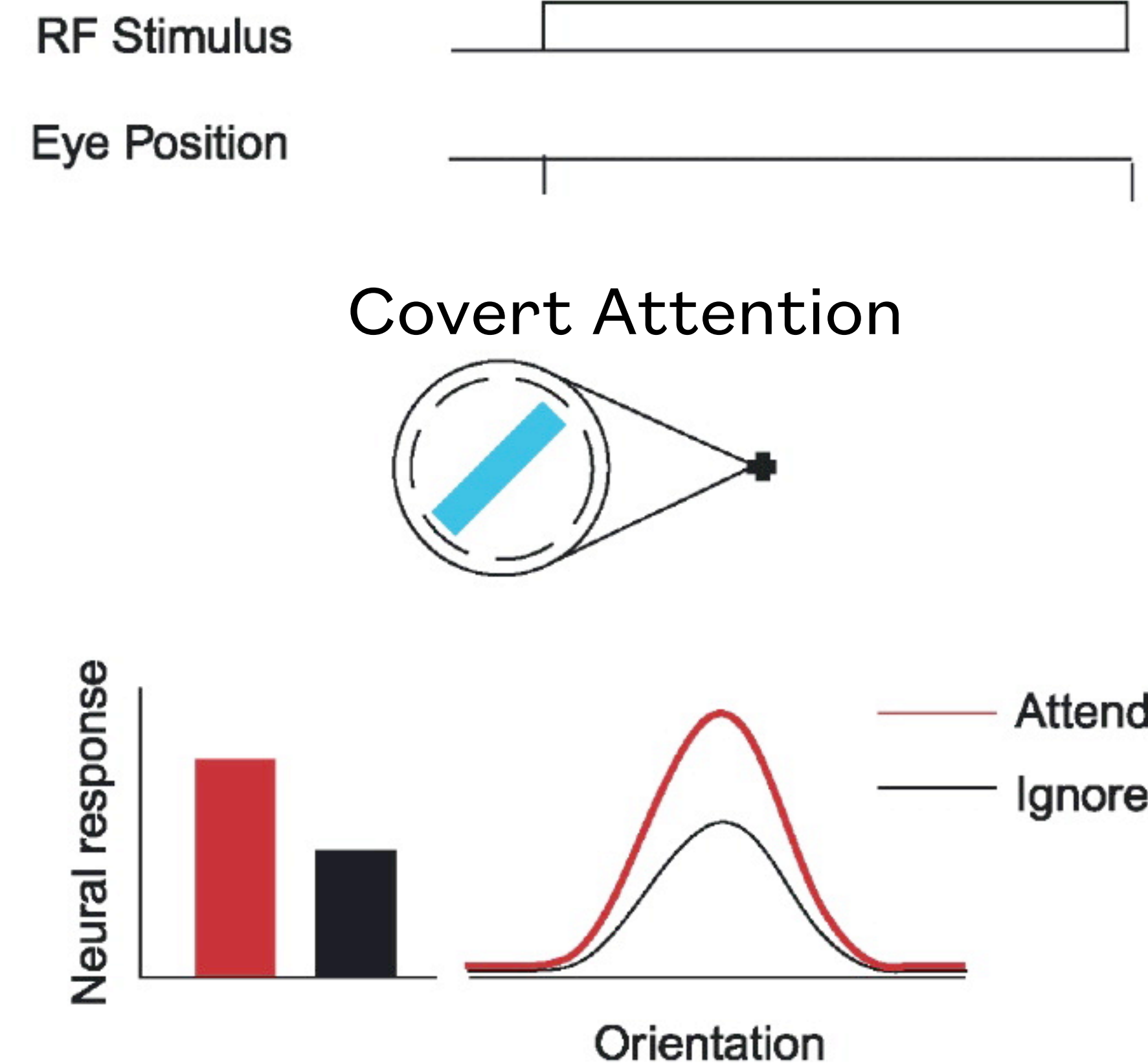
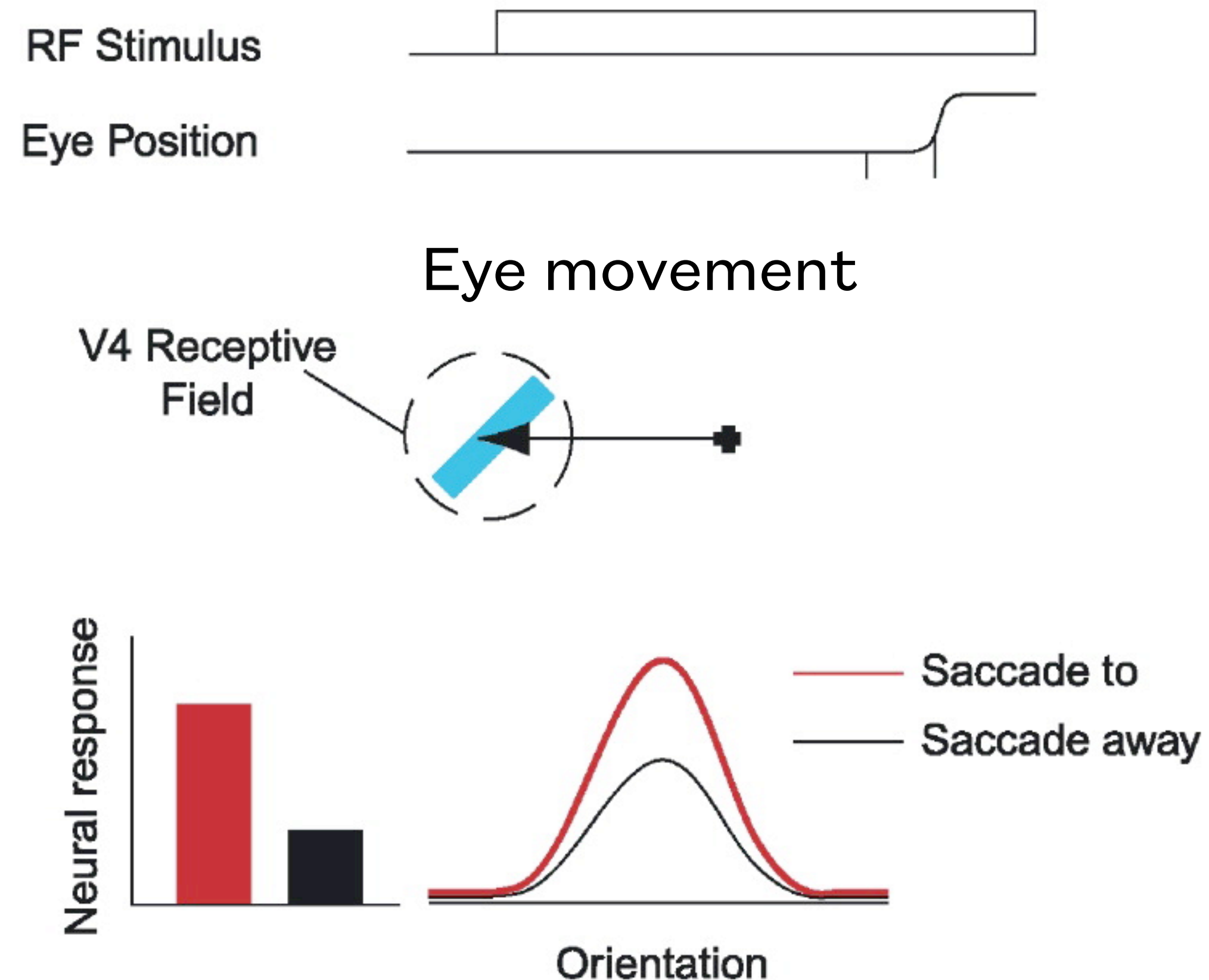
Eye movement network



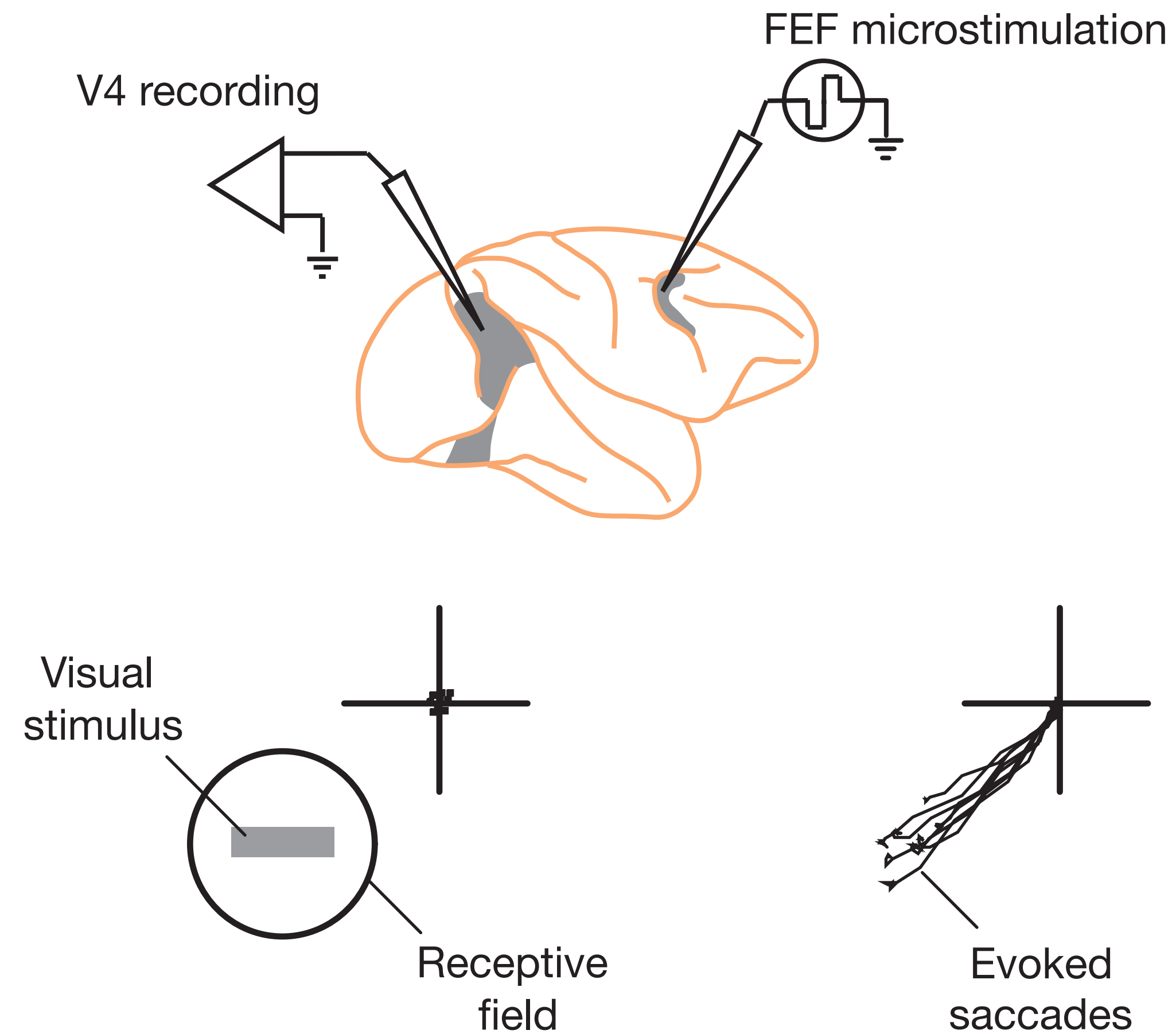
Saccade



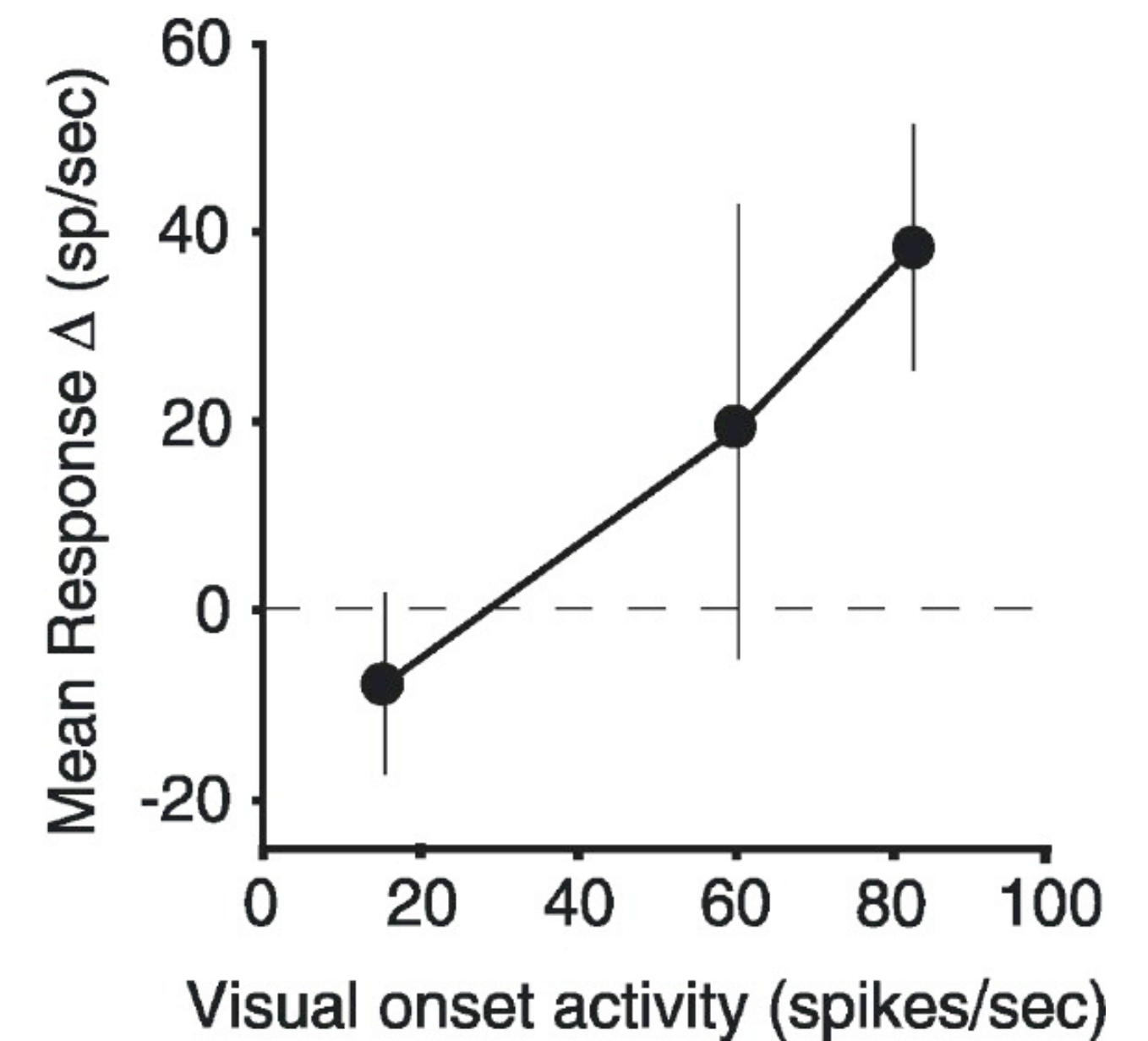
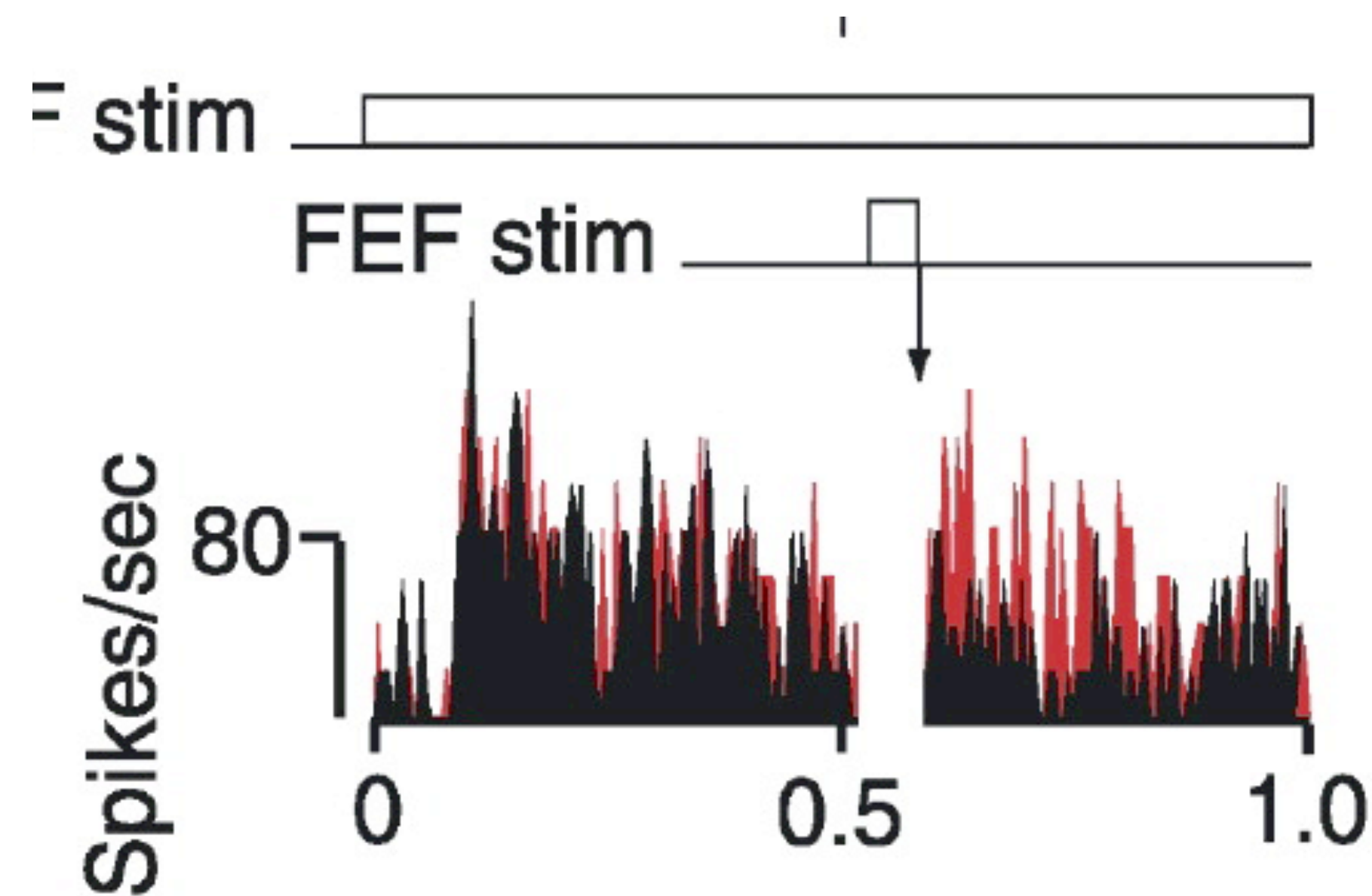
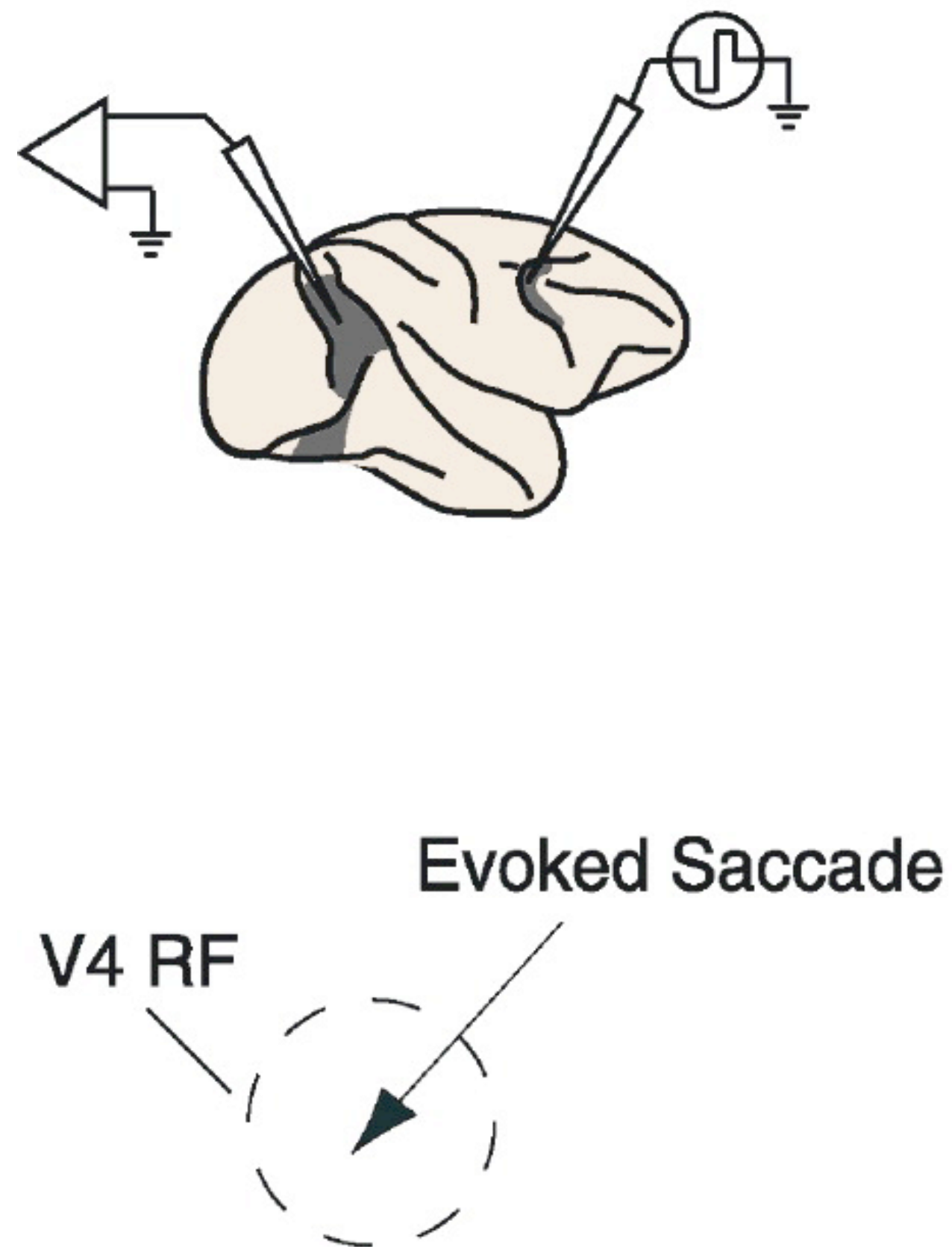
Eye movements & attention both modulate visual responses



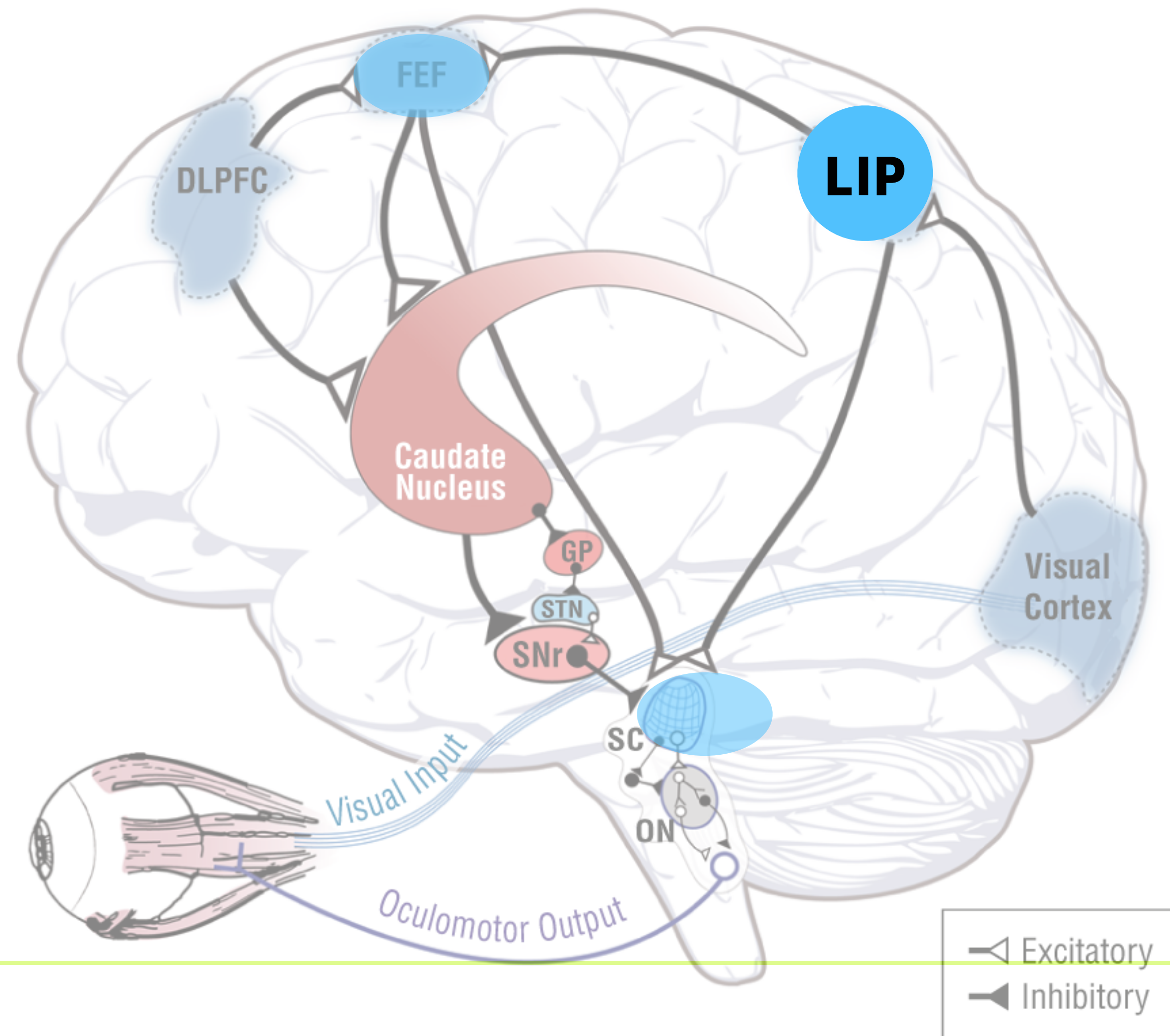
FEF Mediates Attention



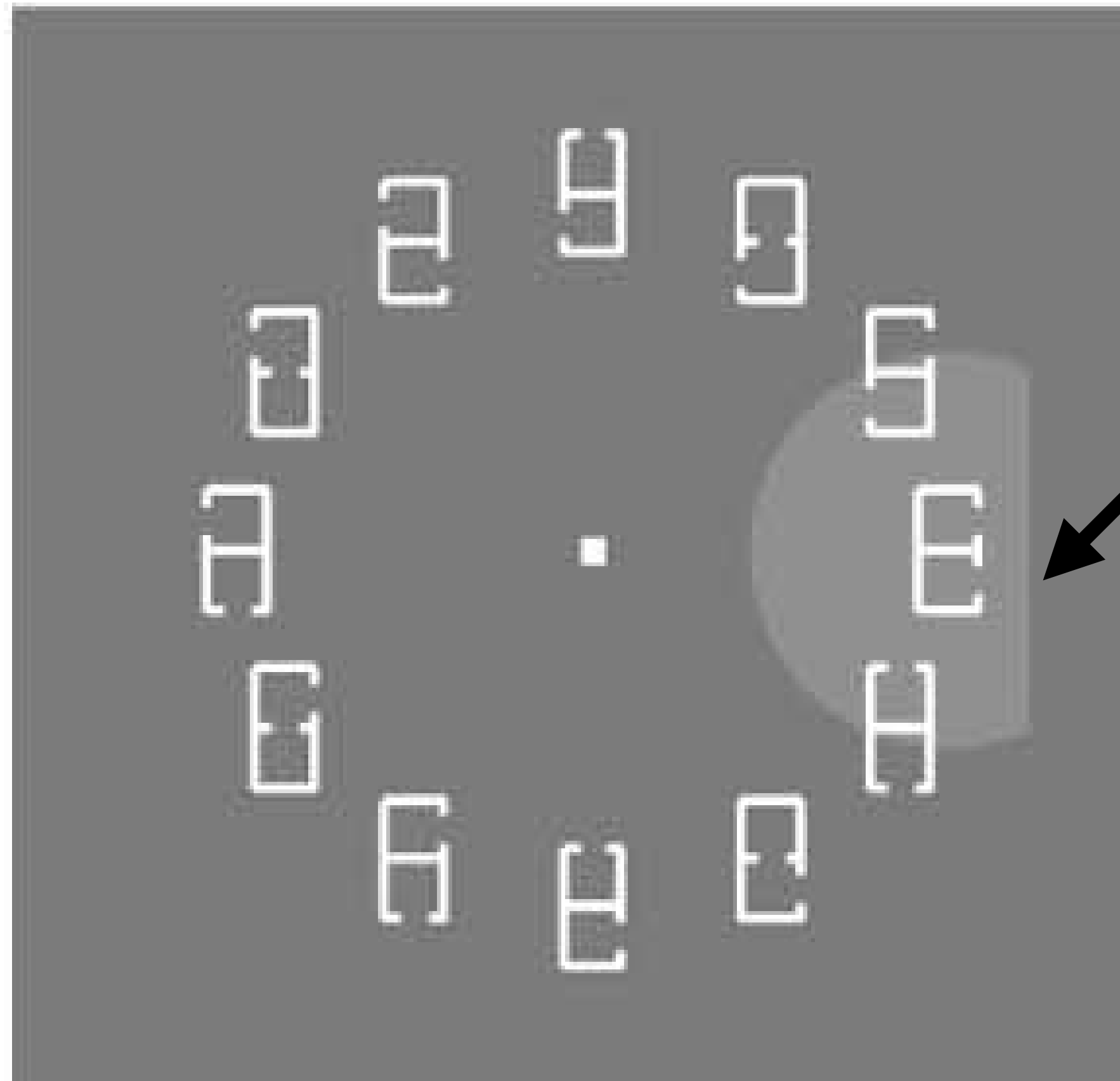
FEF μ -stimulation modulates V4 responses



LIP



LIP: Bottom-Up Attention



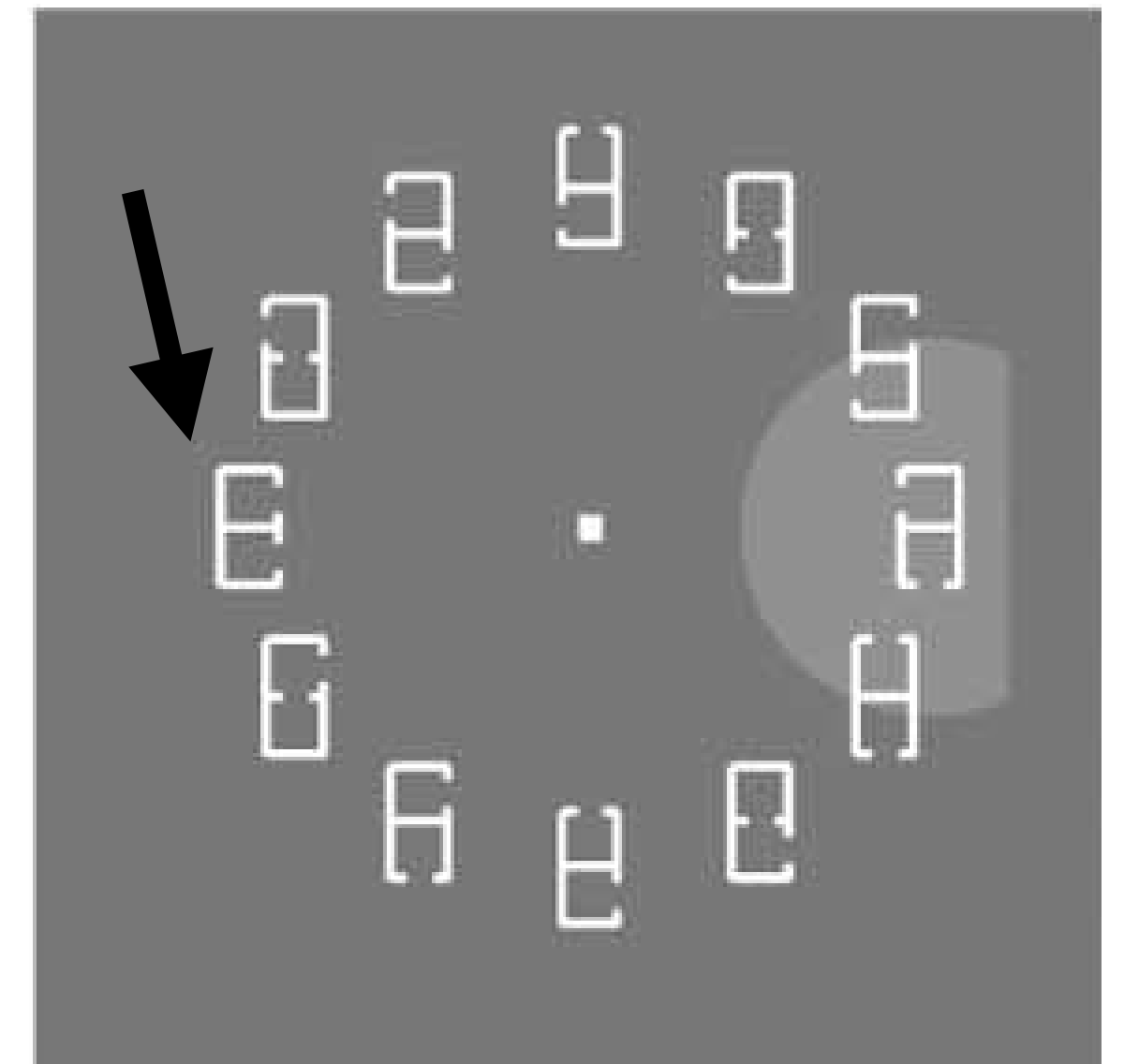
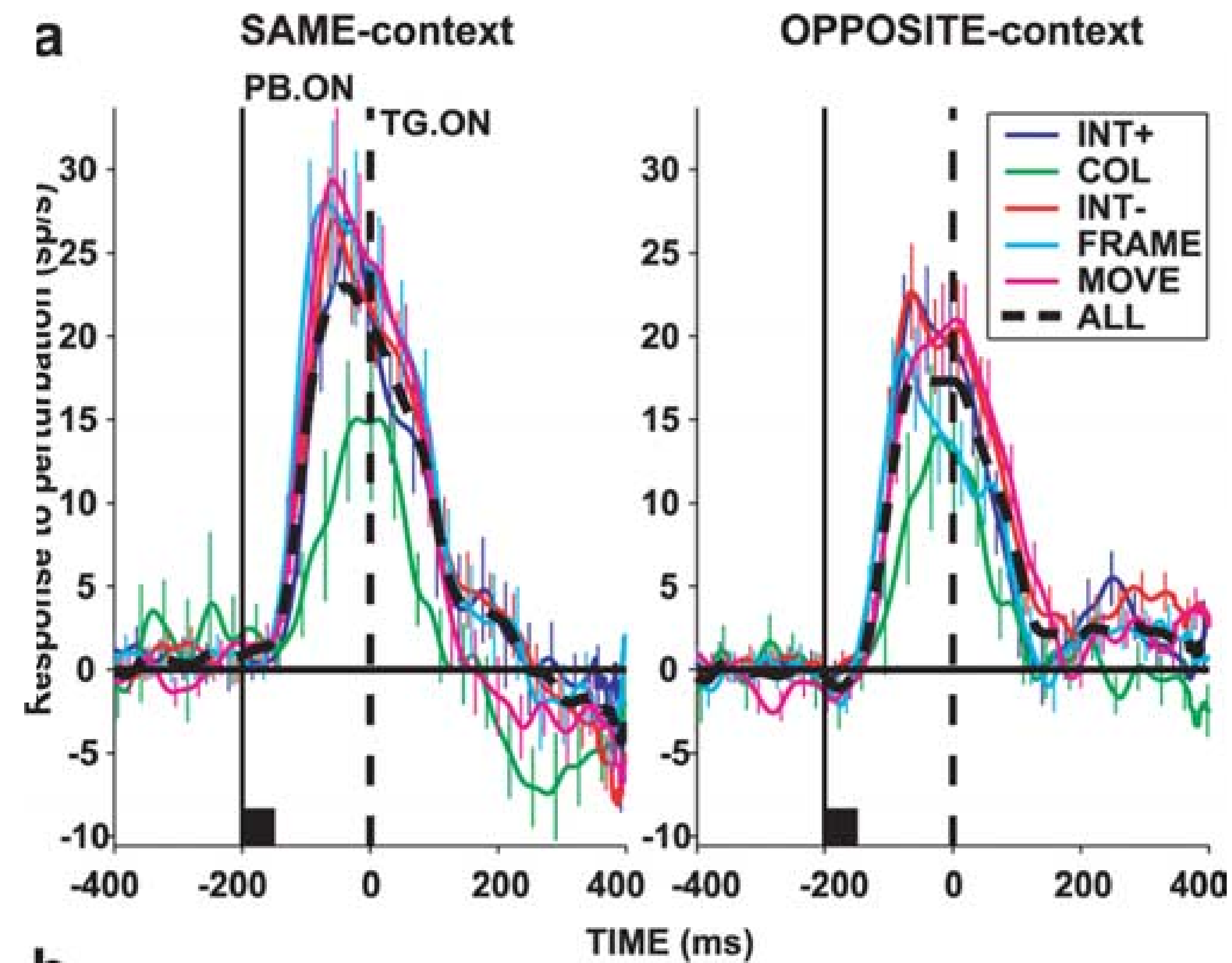
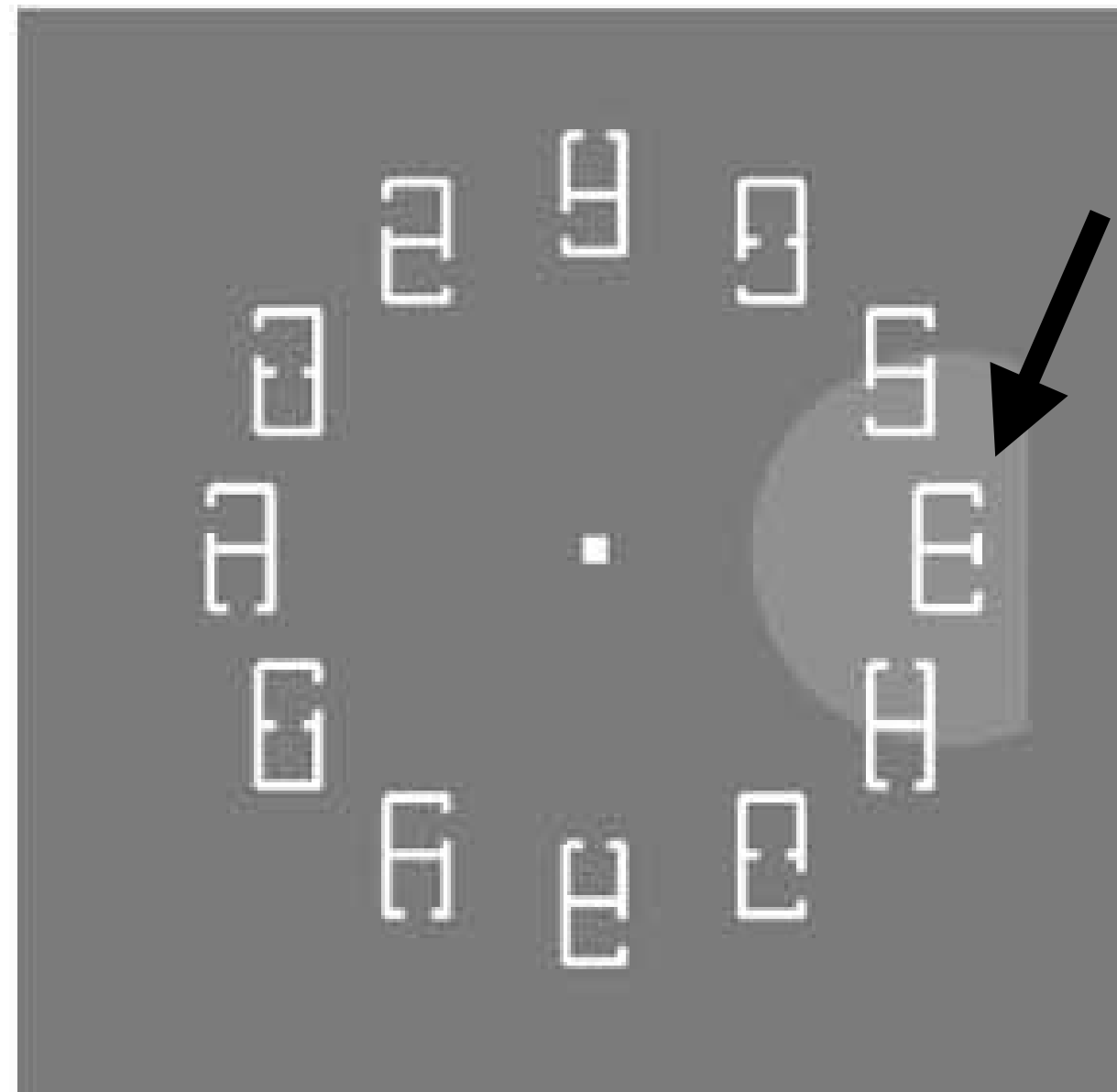
Different Color

Different Intensity (+&-)

Add a frame

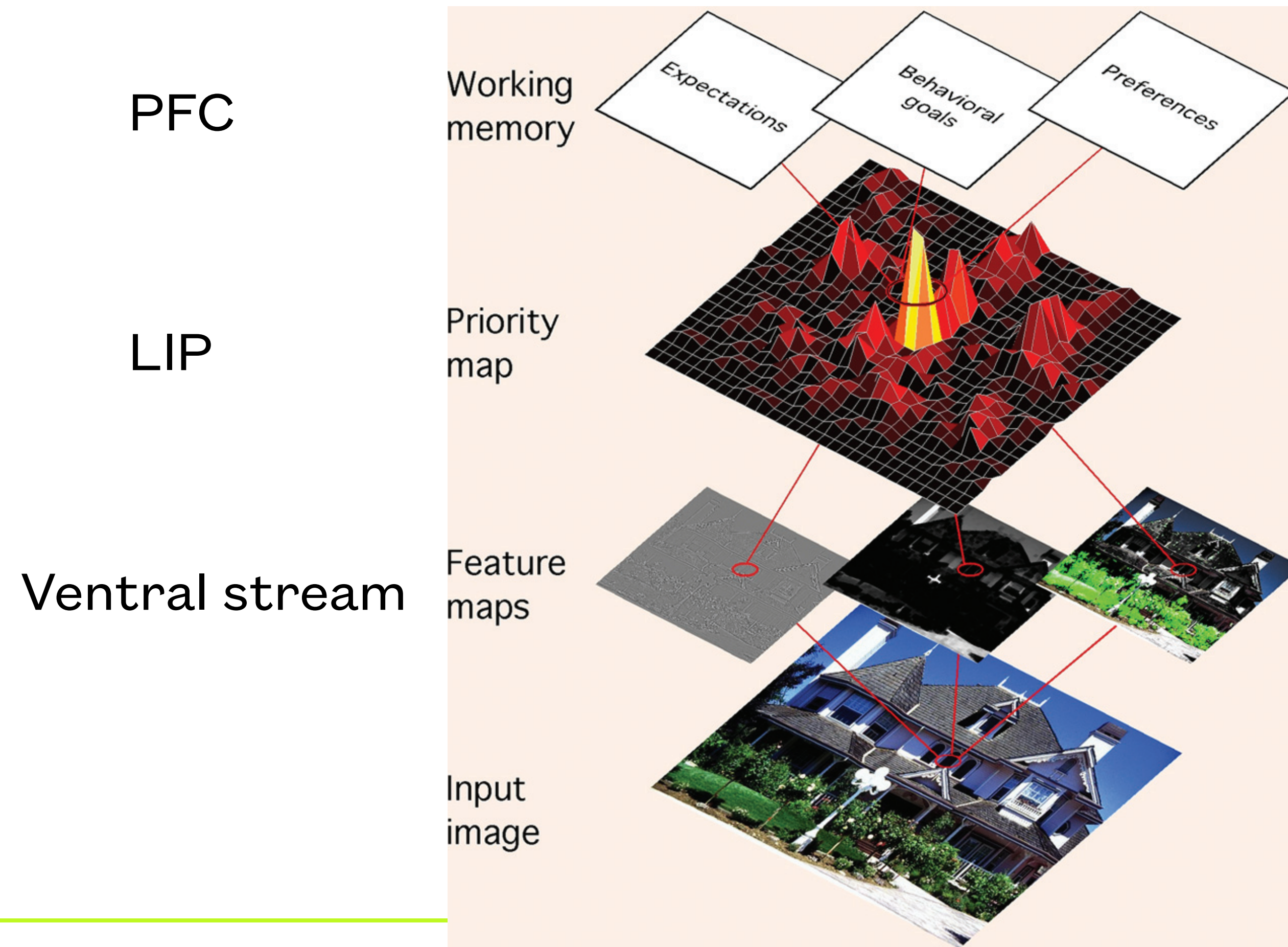
Moves a little

LIP: Bottom-Up Attention

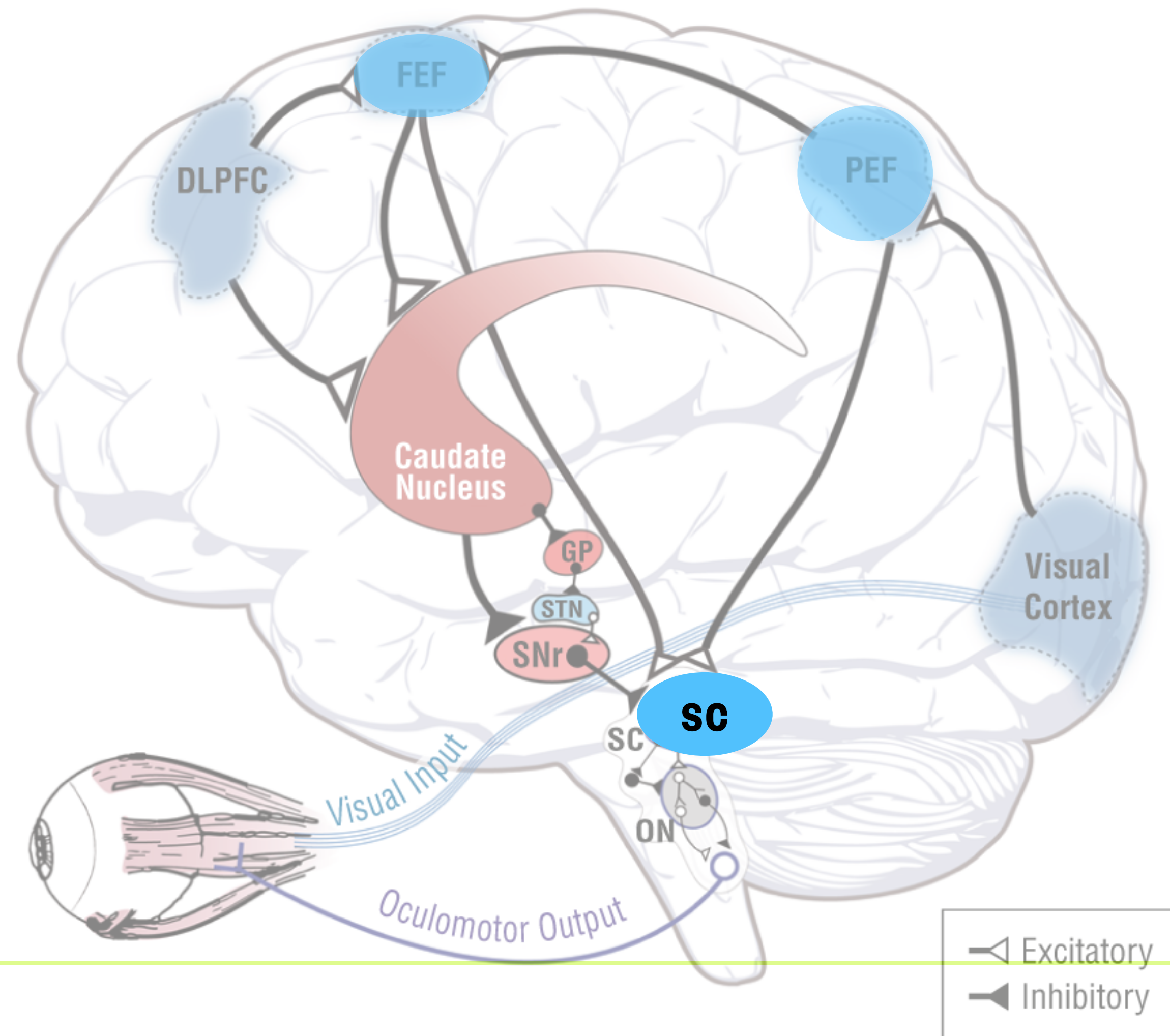


LIP: Priority Map

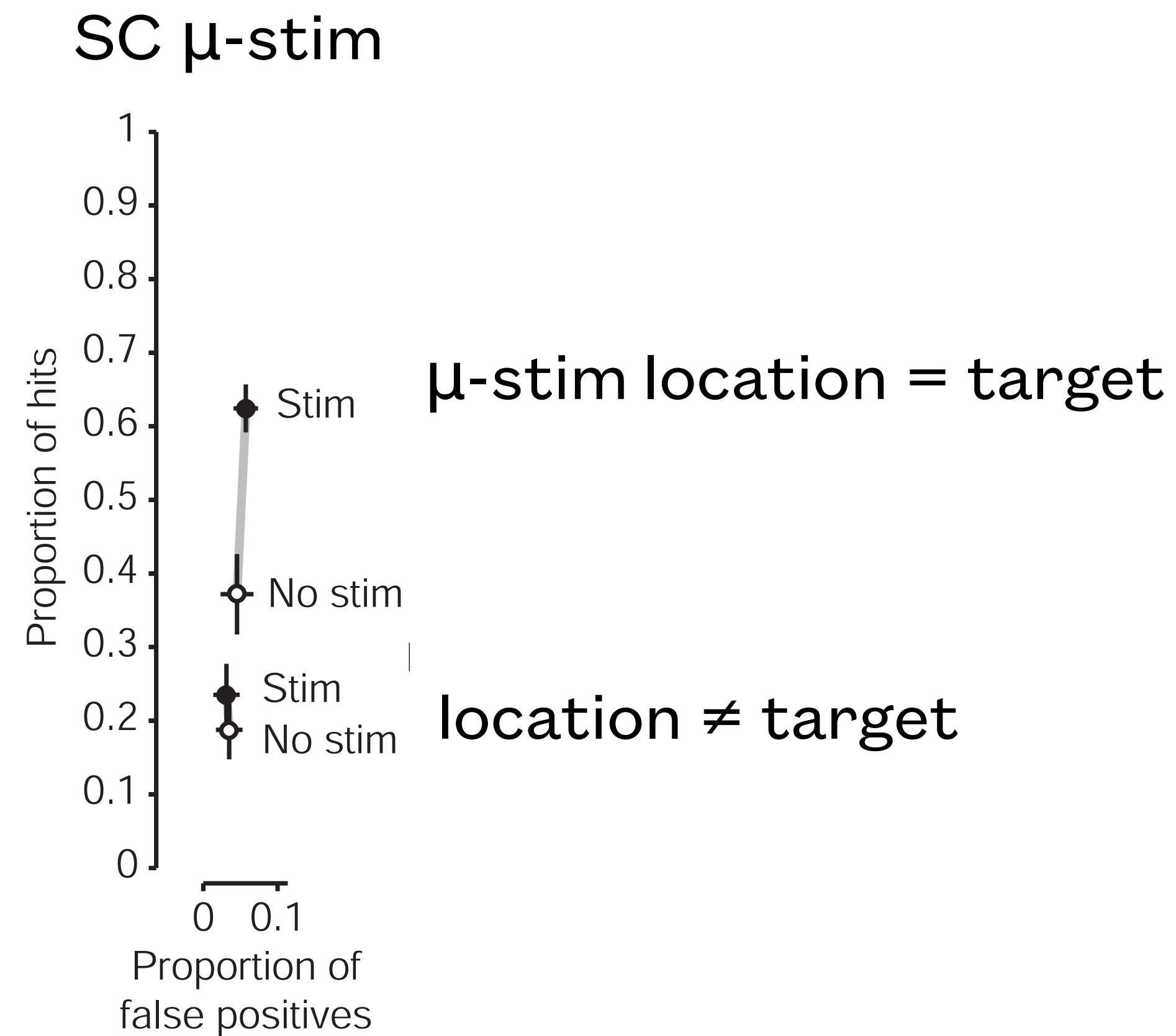
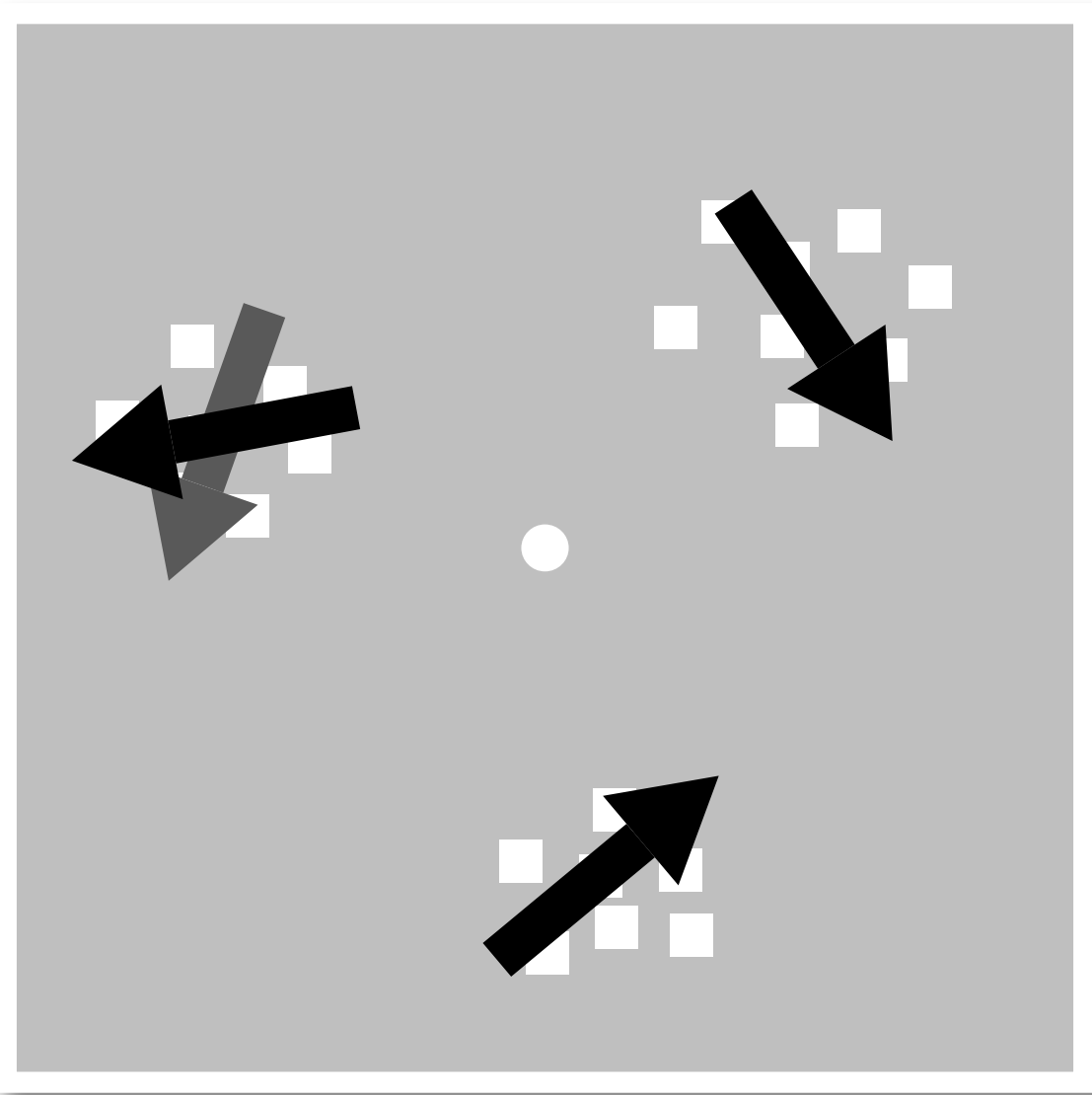
Bisley and Goldberg, 2010



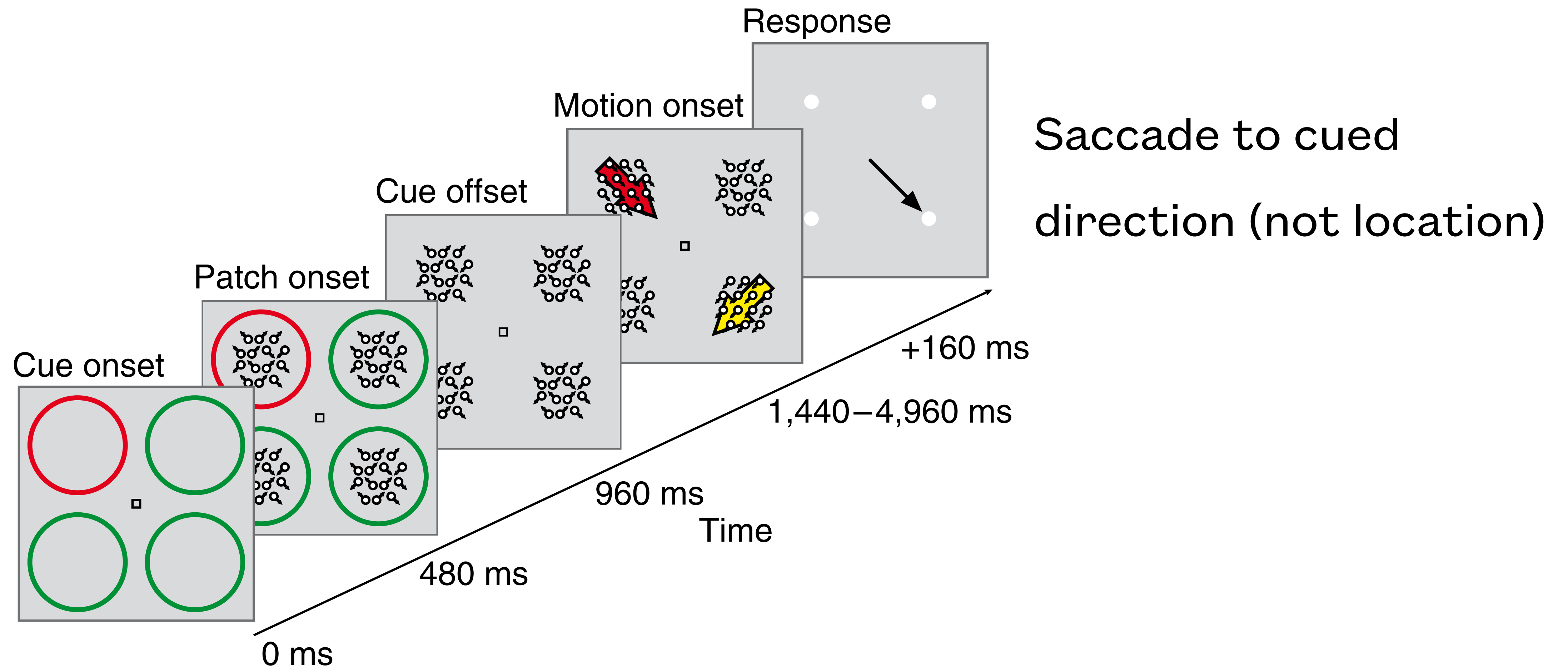
Superior Colliculus



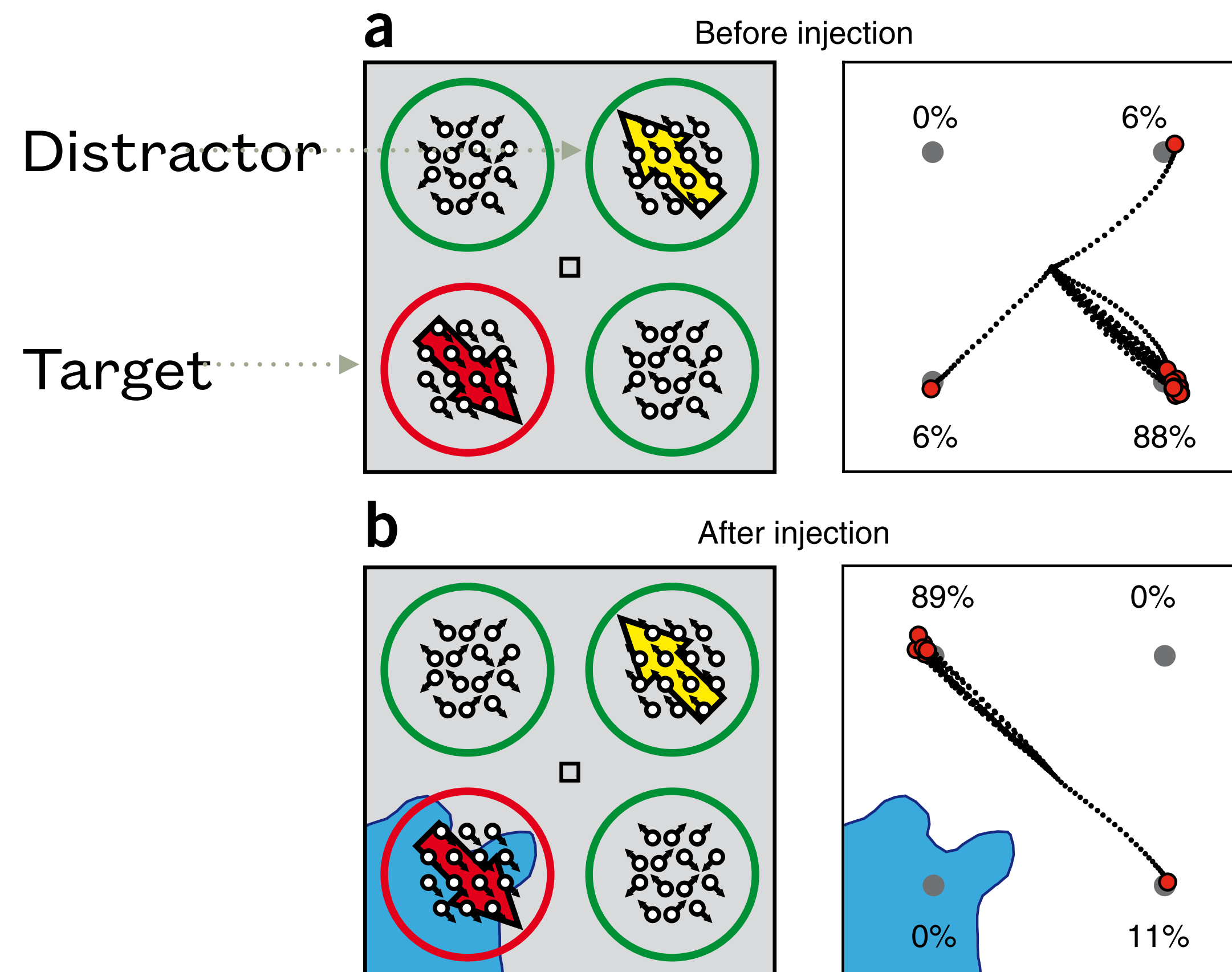
Superior Colliculus μ -stimulation



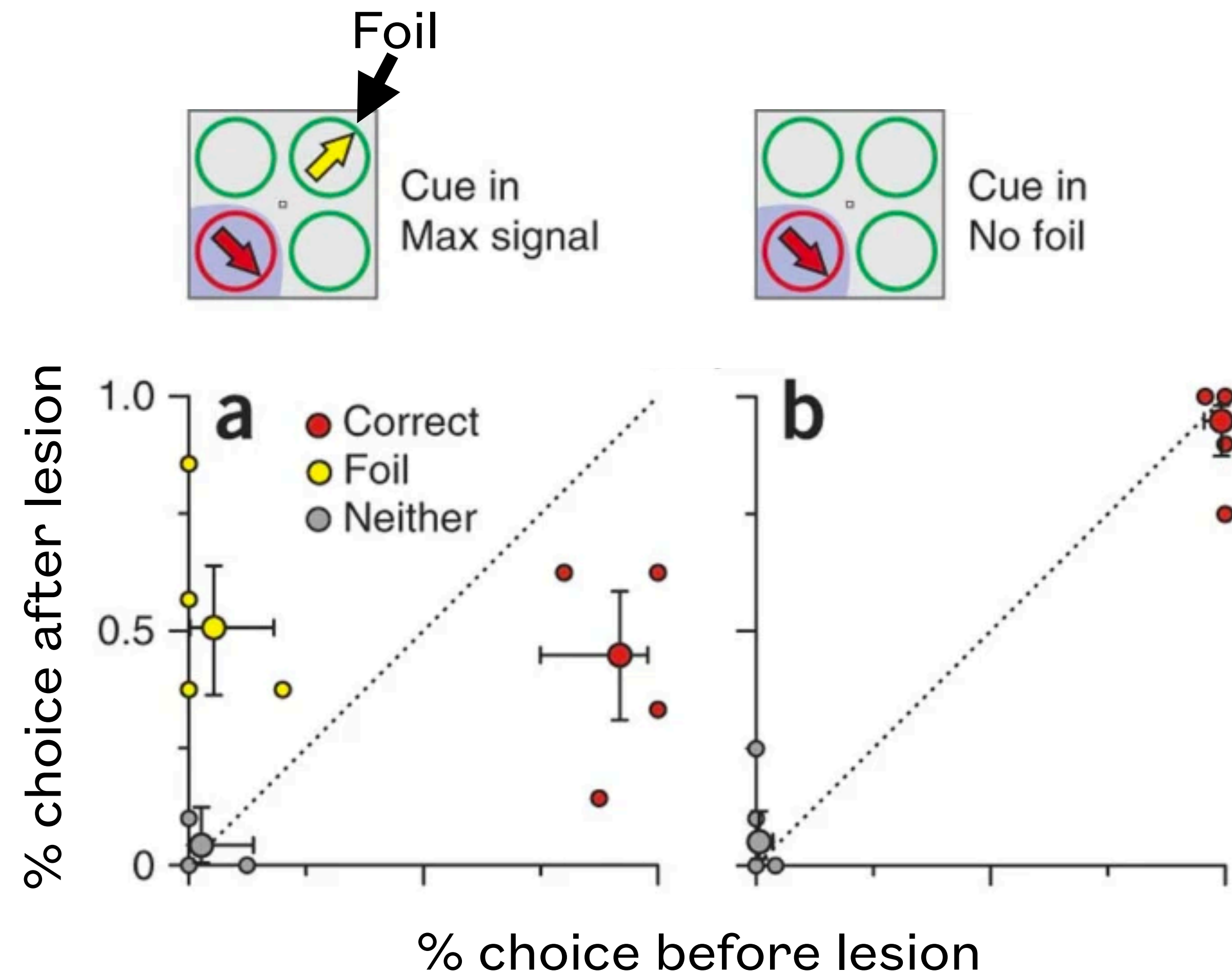
Superior Colliculus Inactivation



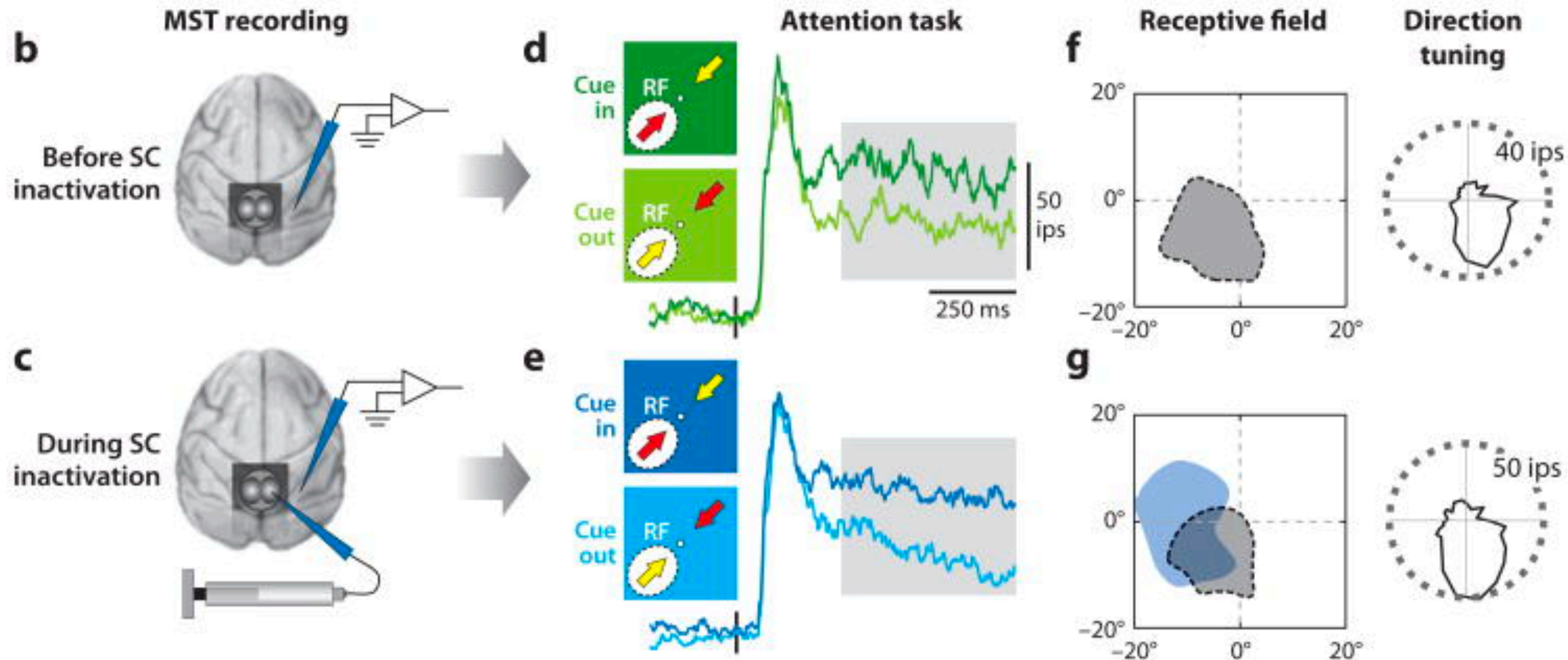
Superior Colliculus Inactivation



Superior Colliculus Inactivation **Not** Explained by Sensory Impairments



SC inactivation does **not** affect attention modulations in visual cortex



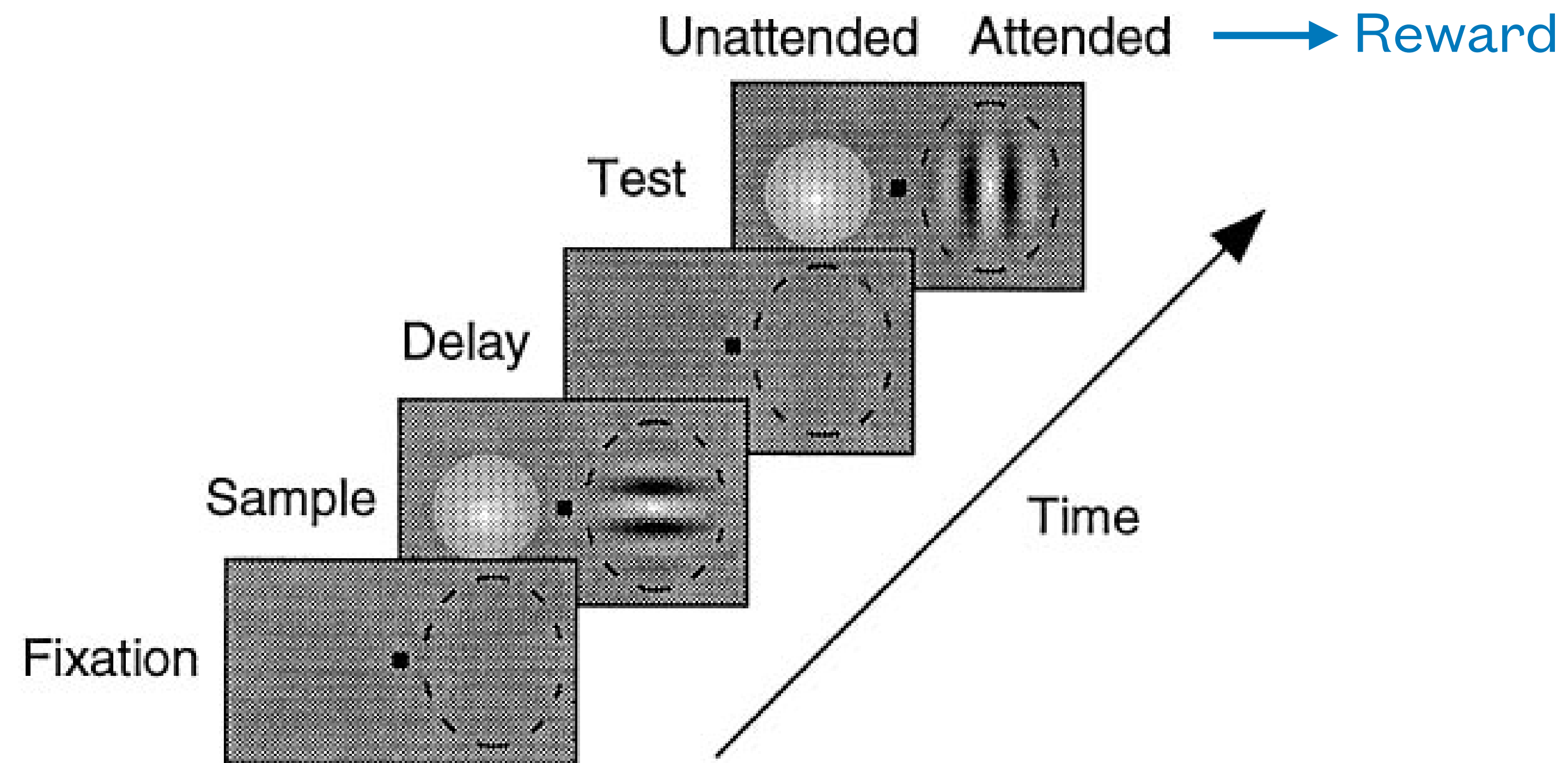
-
- ❖ Overview
 - ❖ Visual Attention
 - ❖ Attention Modulation of Visual Processing
 - ❖ Top-down vs Bottom-up
 - ❖ **And more...**
-



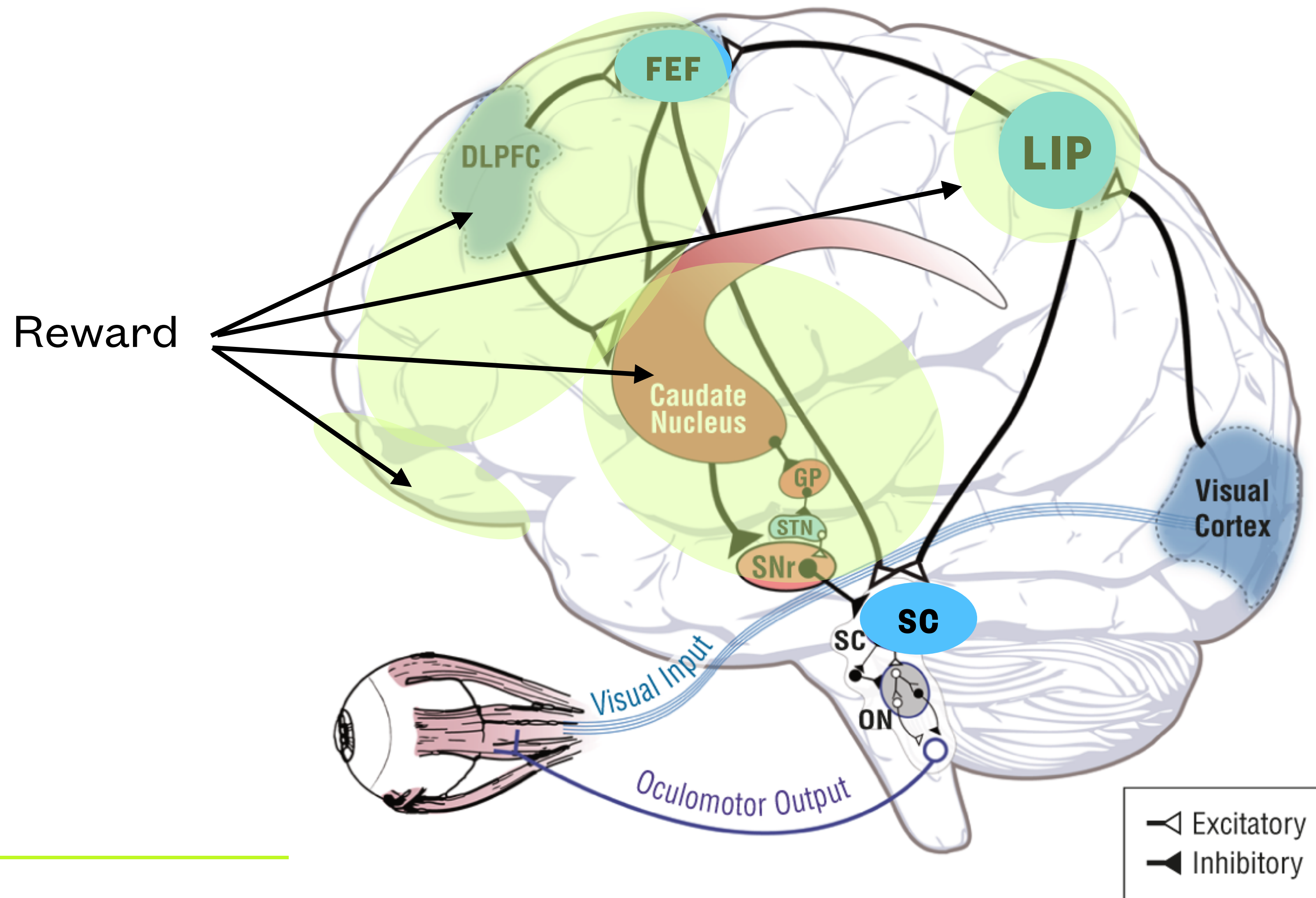
- ❖ Overview
- ❖ Visual Attention
- ❖ Attention Modulation of Visual Processing
- ❖ Top-down vs Bottom-up
- ❖ **And more...**
 - ❖ **Reward**
 - ❖ **Working Memory**
 - ❖ **Decision Making**
 - ❖ **Consciousness**



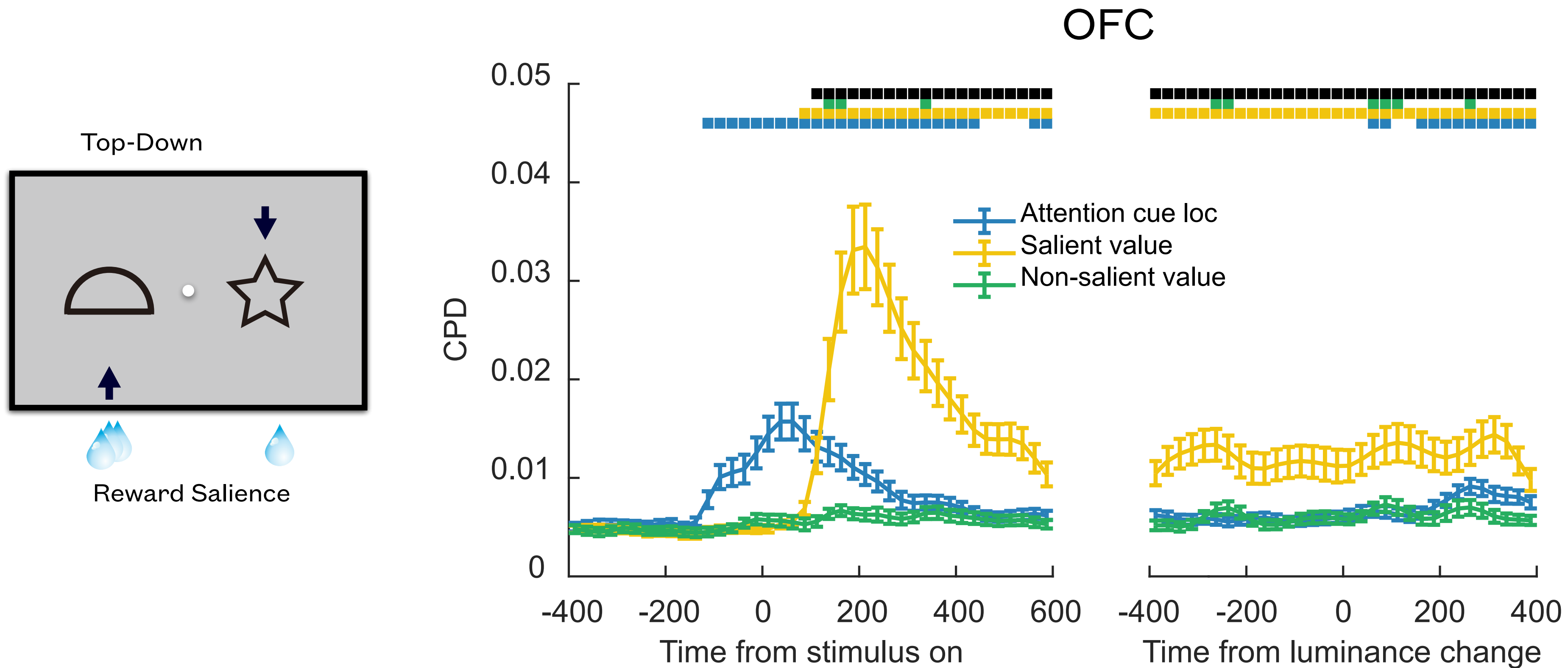
Attention & Reward



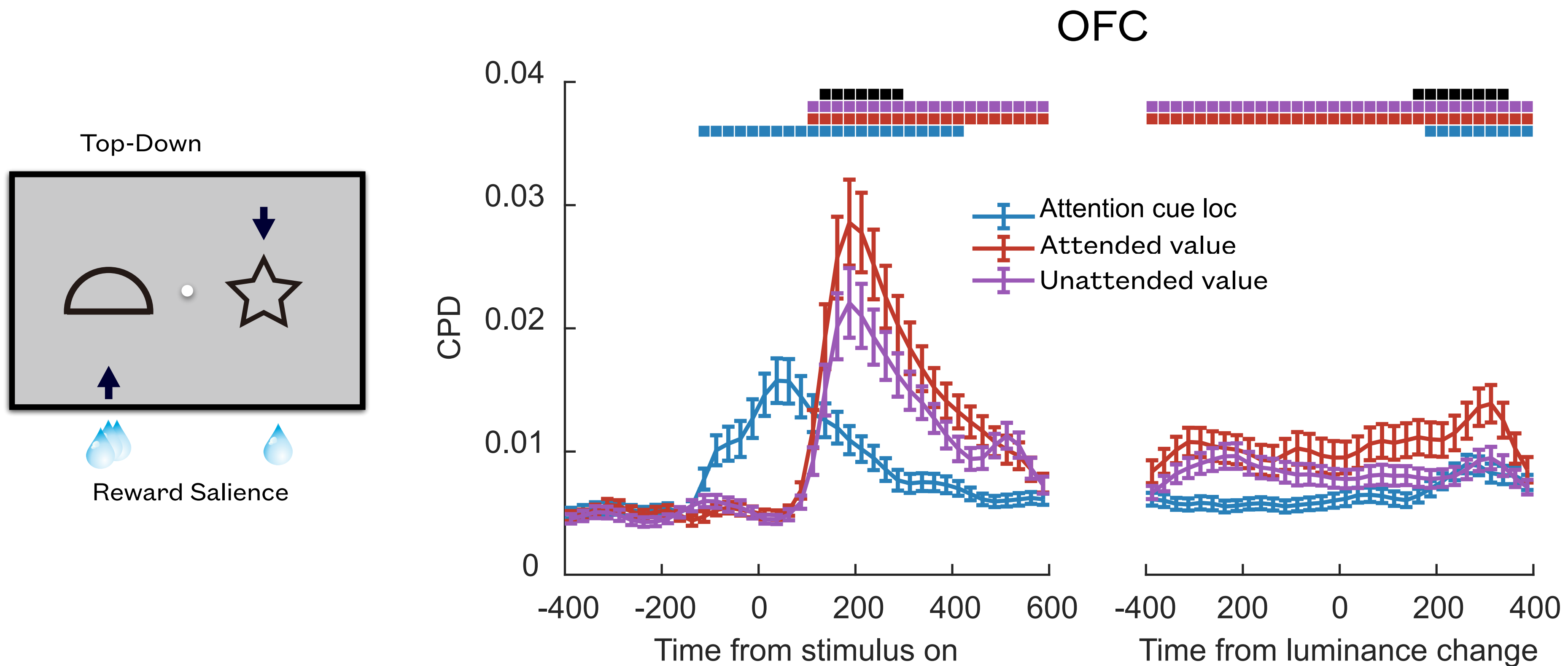
Attention & Reward



Attention & value signals are distinctly represented in PFC



Attention & value signals are distinctly represented in PFC



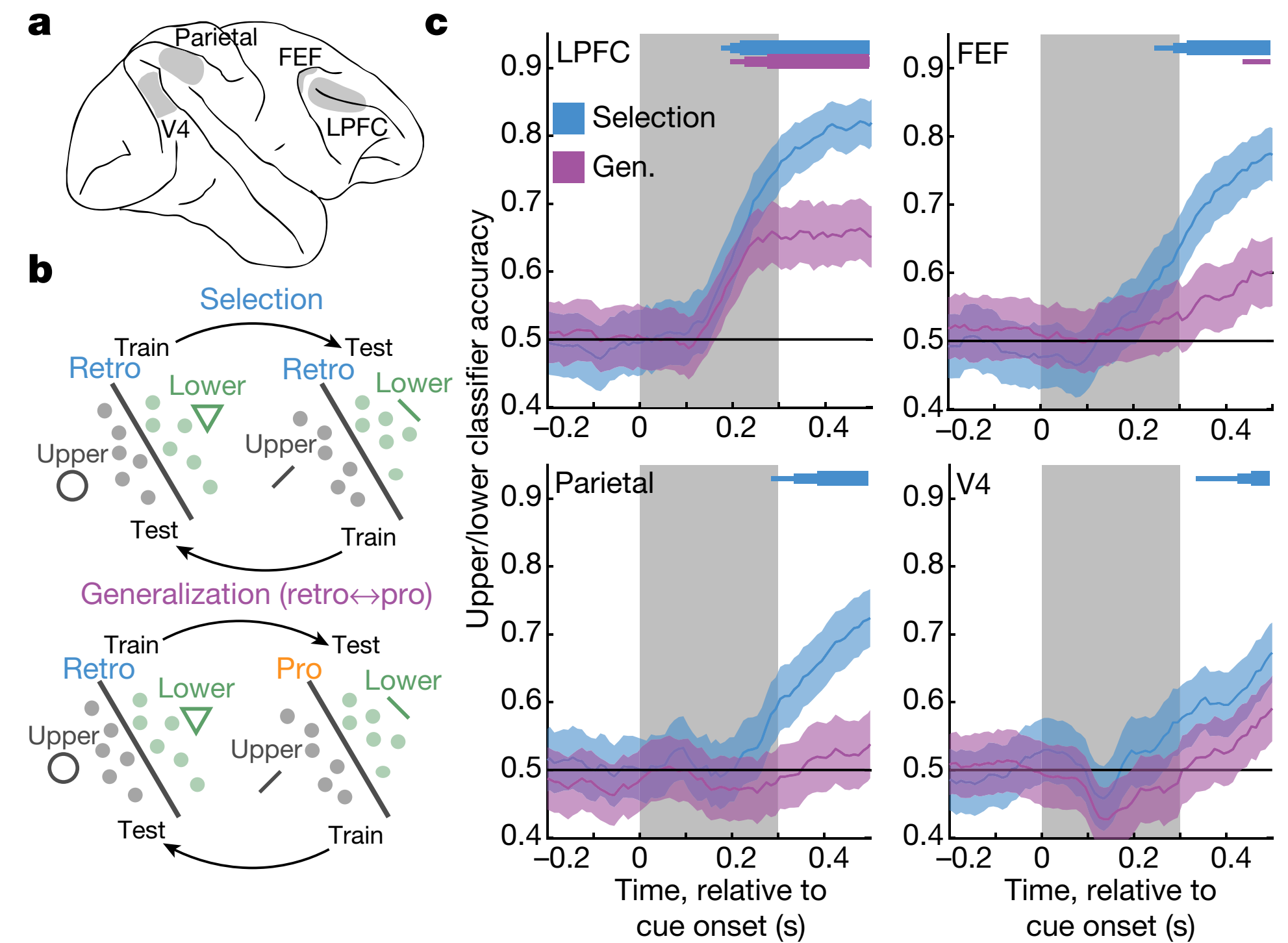
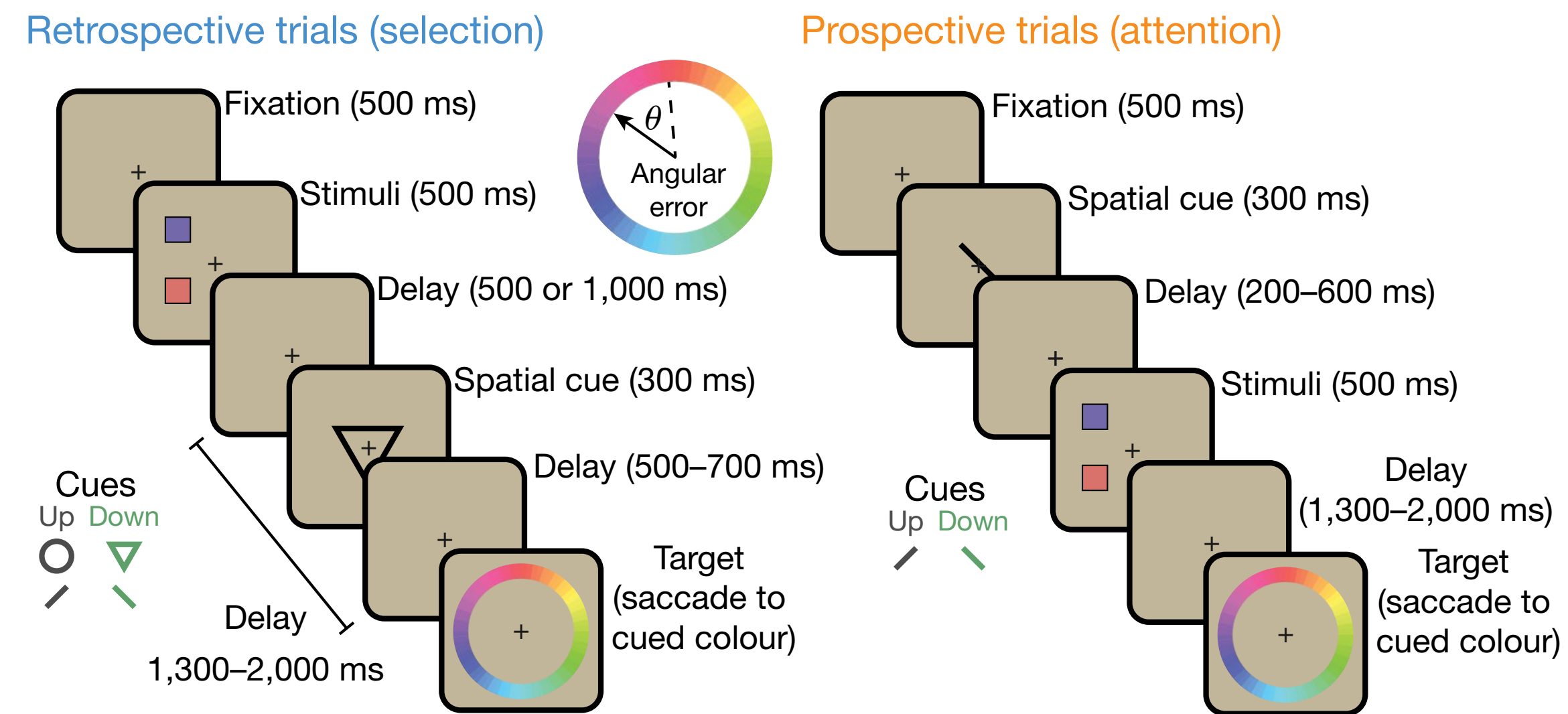
Attention & Working Memory



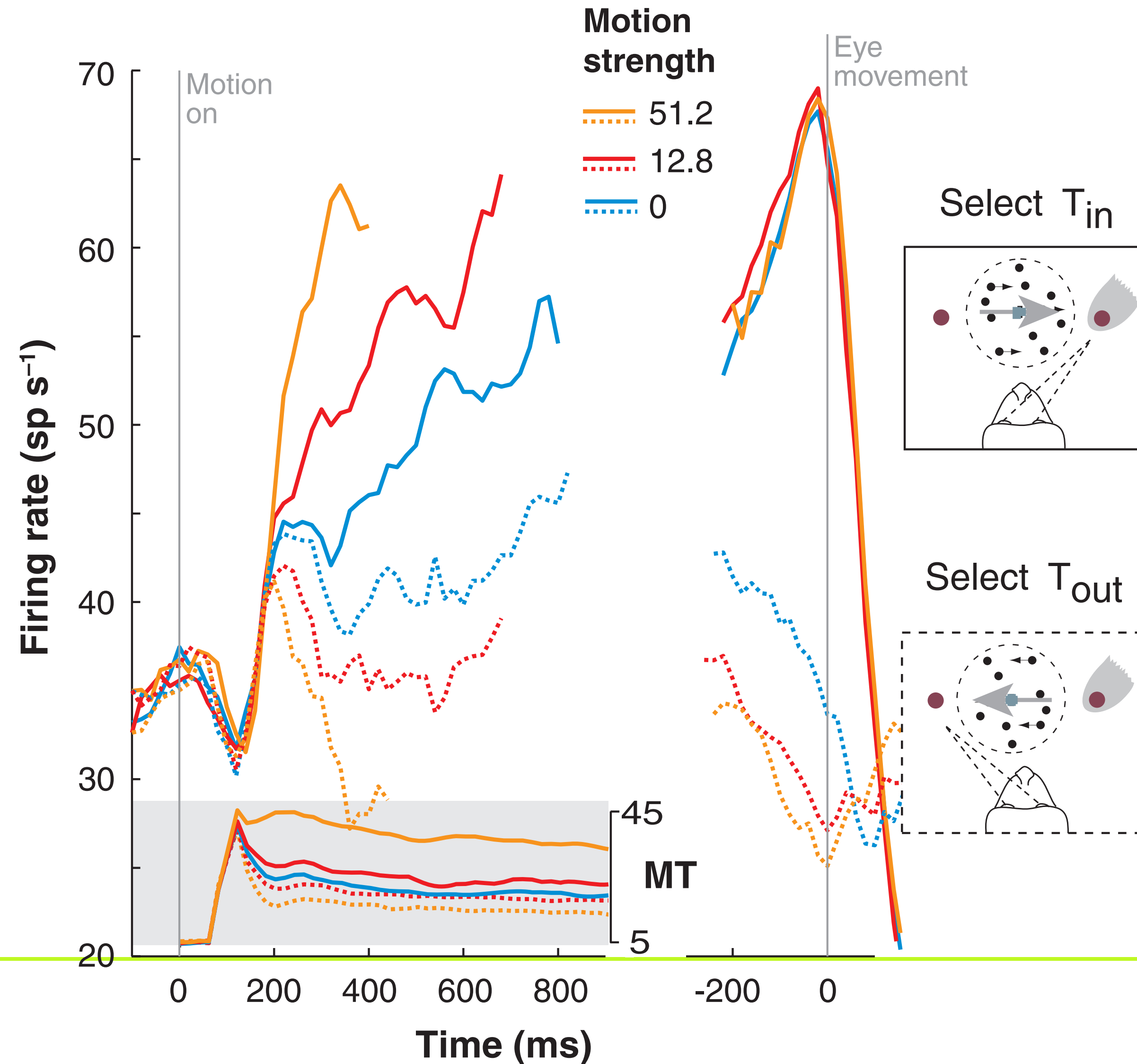
Working Memory

- By virtue of holding a selected subset of all available representations in memory, WM is by definition a form of attention.
- The selection of information to be held in WM is a form of controlled attention
- Control of attention relies on WM.

Attention & Working Memory

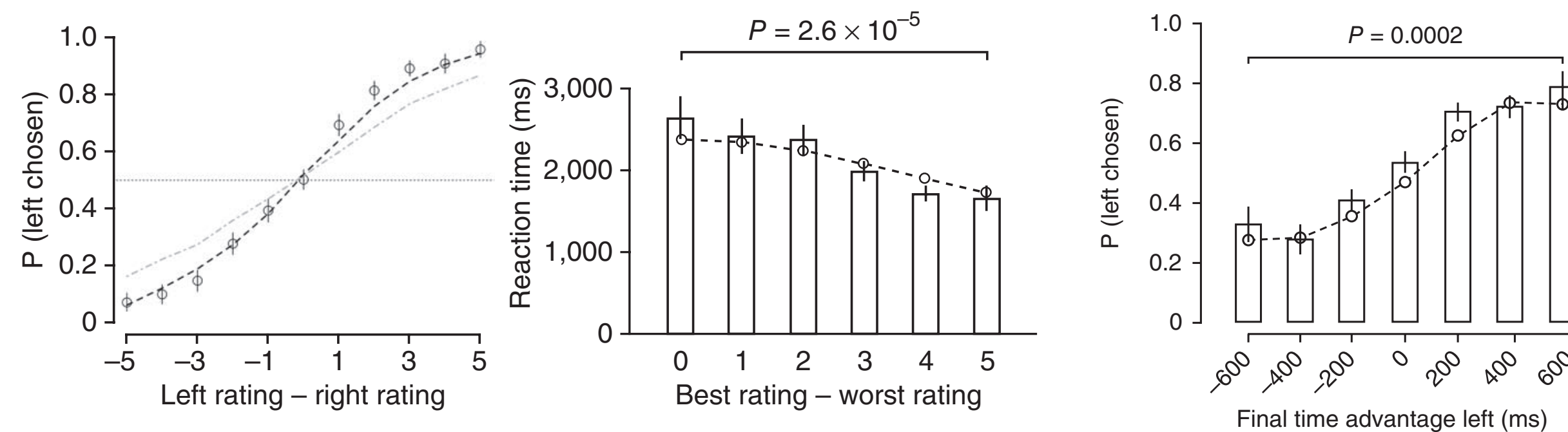
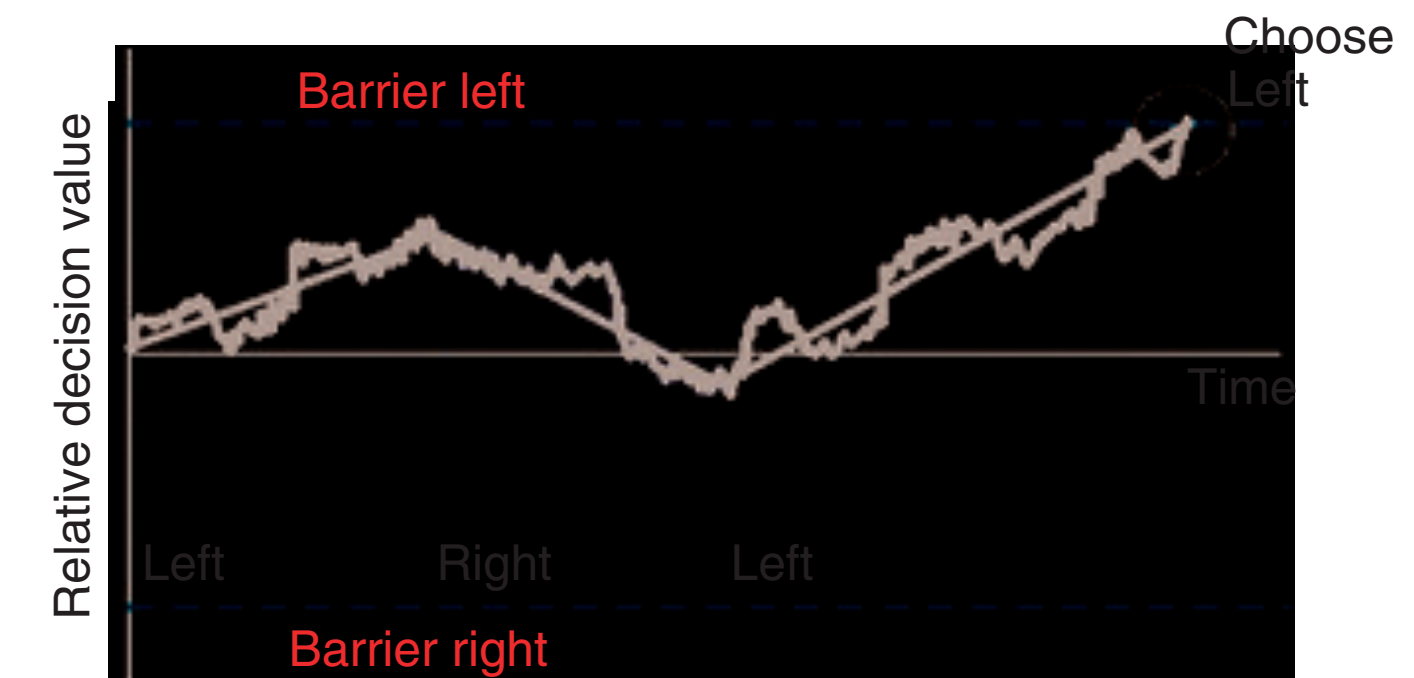


Attention & Decision Making



Attention may affect accumulation

Value-based decision making

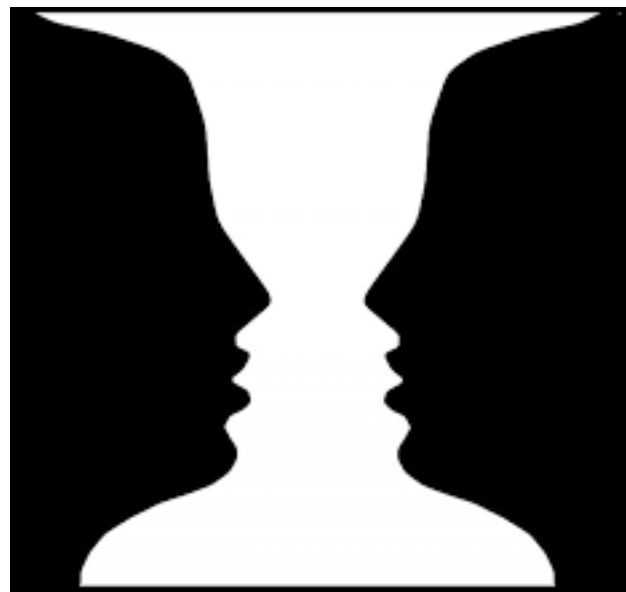


$$\Delta DV = d(r_{fix} - \theta \cdot r_{nonfix}) + \epsilon$$

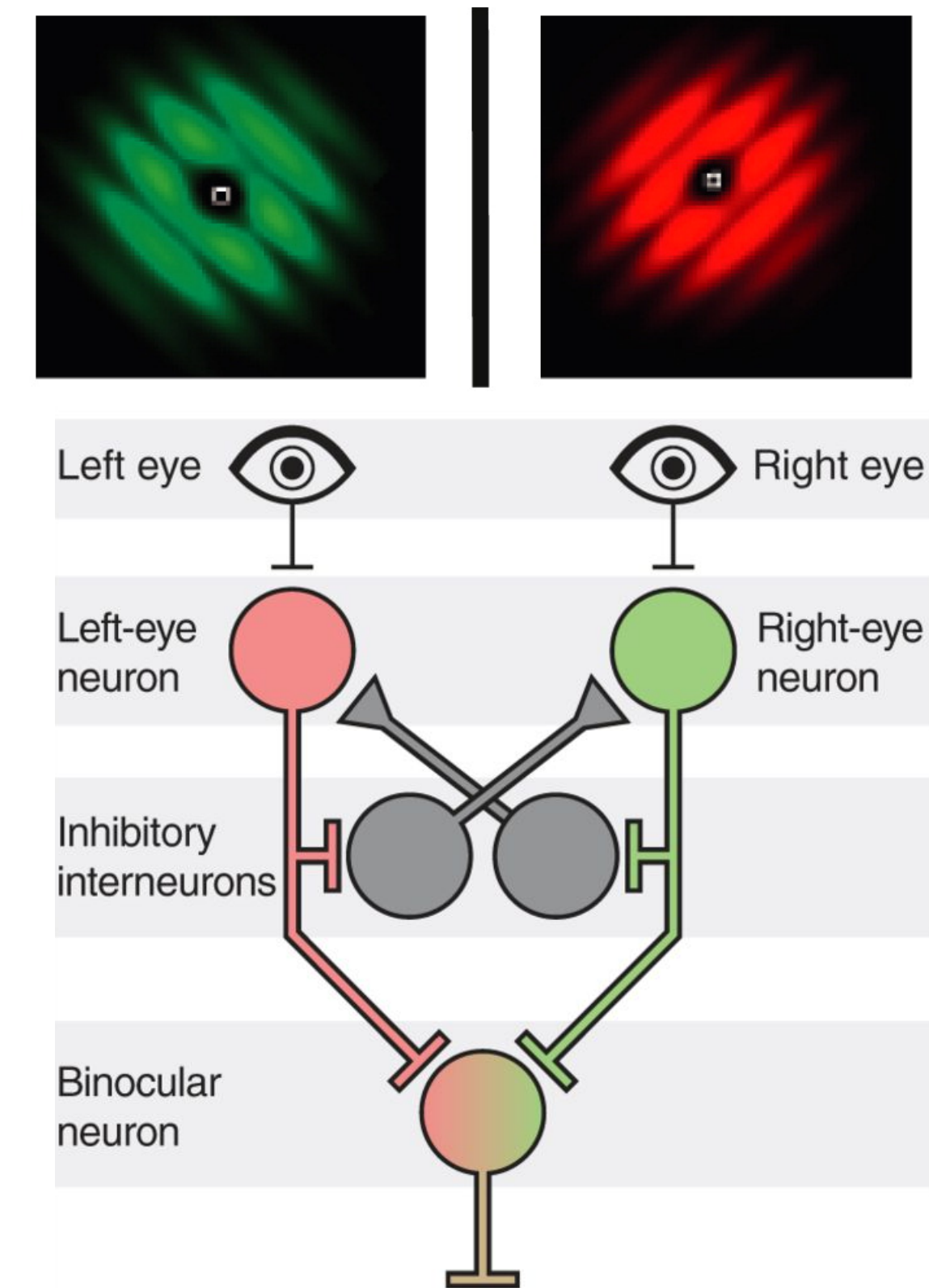
Diagram illustrating the components of the drift-diffusion model equation:

- ΔDV : Decision Variable
- d : Drift rate
- r_{fix} : Reward Fixation side
- θ : Attention Bias
- r_{nonfix} : Reward Non-fixation side
- ϵ : Noise

Attention and (Visual) Consciousness



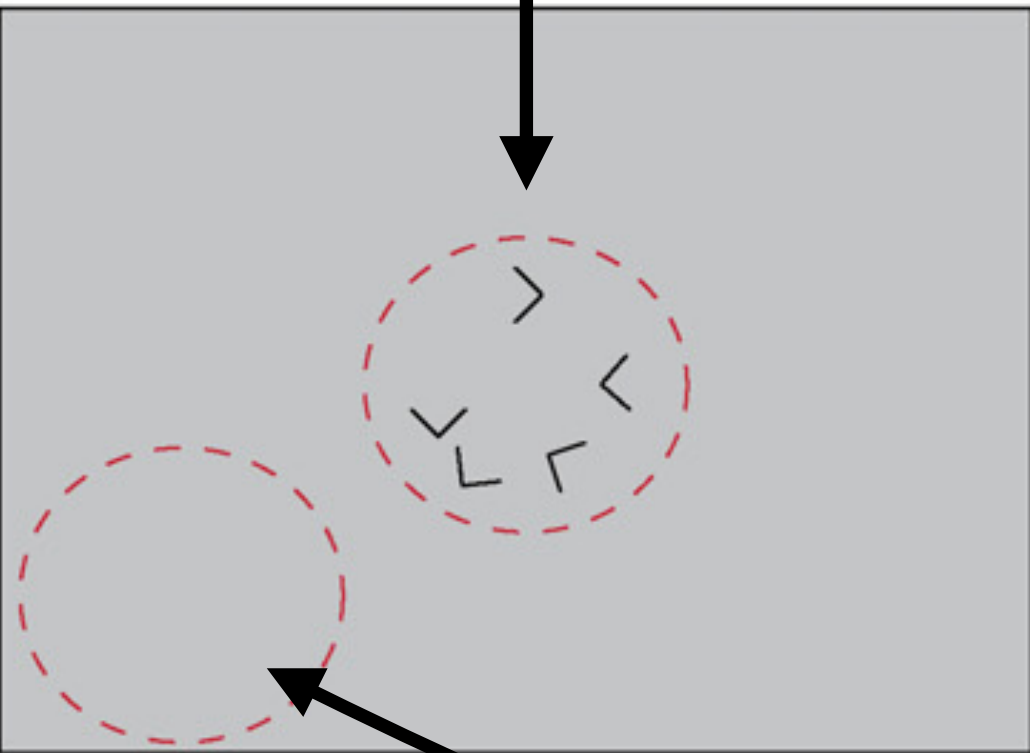
Bi-stable stimulus



Binocular rivalry

Independent manipulations of attention and consciousness

Difficult central task



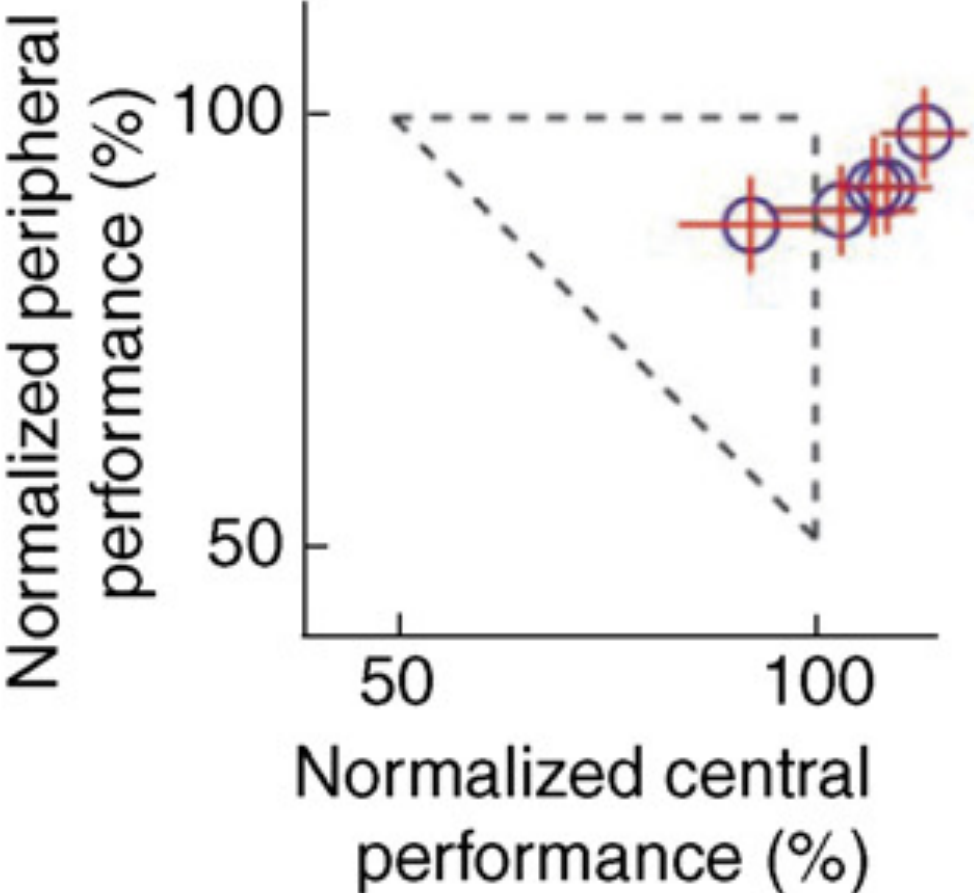
Target: animal



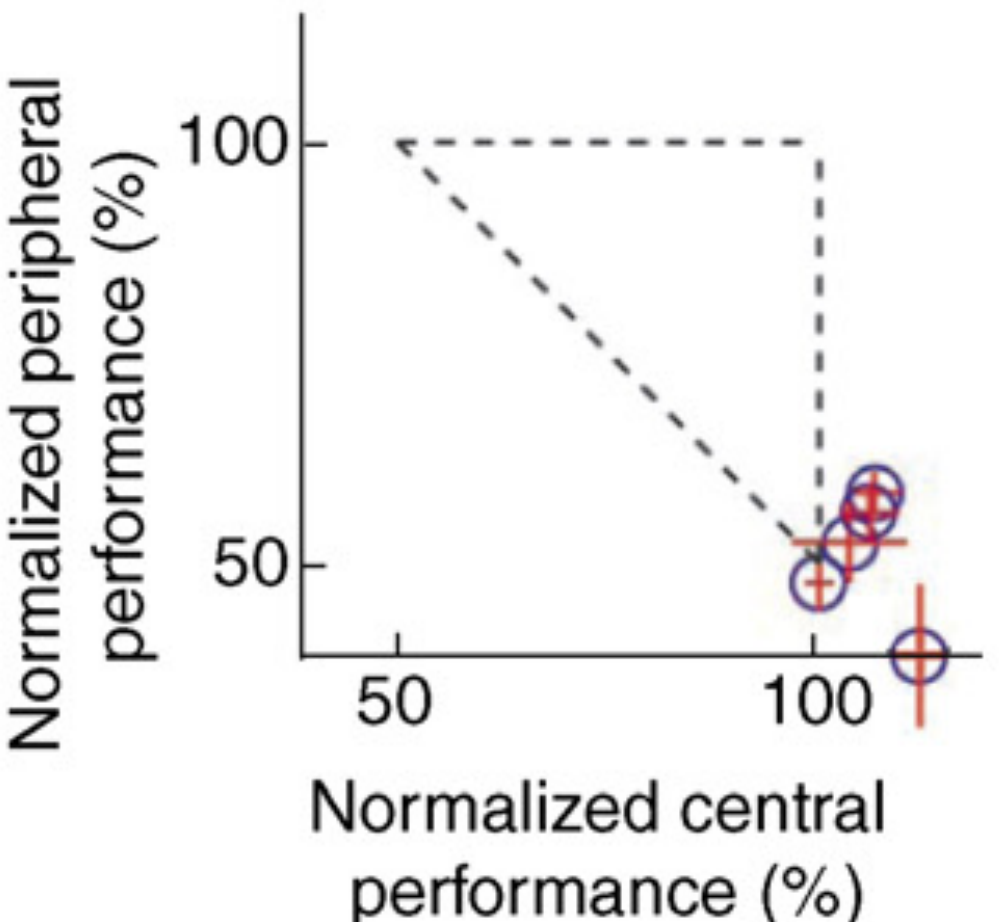
Distractor



vs



vs



Peripheral task

Summary

- Different types of attention
- Posner paradigm
- How attention affects the visual processing in the brain
 - Gain/Contrast Modulation
 - Noise correlation
 - Normalization model
- How attention is generated in the brain
 - Eye movements, FEF/LIP/SC
 - Attention may be a combination of distinct processes
- Reward, working memory, decision making, and consciousness

